Physica bk4

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1 On GlossySnake

A LaTeX Proof-of-Concept of the GlossySnake project. By the nature of the machine-glossing, the inaccurate and malformatted glossing is expected.

2 Physica bk4

PHYSICA Bk.IV

PHYSICS Book IV

Κεφάλαιο 1

Chapter 1

Ὁμοίως δ' ἀνάγκη καὶ περὶ τόπου τὸν φυσικὸν ὥσπερ καὶ περὶ

Likewise but necessity and about place the physical as and about

ἀπείρου γνωρίζειν, εἰ ἔστιν ἢ μή, καὶ πῶς ἔστι, καὶ τί ἐστιν.

infinity to know, whether it is or not, and how it is, and what it is.

τά τε γὰρ ὄντα πάντες ὑπολαμβάνουσιν εἶναί που (τὸ γὰρ μὴ

The and for beings all assume to be somewhere (the for not

ὂν οὐδαμοῦ εἶναι· ποῦ γάρ ἐστι τραγέλαφος ἢ σφίγξ;) καὶ τῆς

being anywhere to be; where for it is a tragihedra or sphinx;) and of the

κινήσεως ἡ κοινὴ μάλιστα καὶ κυριωτάτη κατὰ τόπον

motion the common especially and most authoritative according to place

ἐστίν, ἣν καλοῦμεν φοράν. ἔχει δὲ πολλὰς ἀπορίας τί ποτ'

it is, which we call motion. It has but many perplexities what exactly

ἐστὶν ὁ τόπος· οὐ γὰρ ταὐτὸν φαίνεται θεωροῦσιν ἐξ ἁπάντων
it is the place; not for the same appears they consider from all
τῶν ὑπαρχόντων. ἔτι δ' οὐδ' ἔχομεν οὐδὲν παρὰ τῶν ἄλλων
the existing. Further but nothing we have nothing beside the others
οὔτε προηπορημένον οὔτε προηυπορημένον περὶ [208b] αὐτοῦ.
nor pre-considered nor previously considered about [208b] of it.
ὅτι μὲν οὖν ἔστιν ὁ τόπος, δοκεῖ δῆλον εἶναι ἐκ τῆς
that indeed therefore there is the place, it seems clear to be from the
ἀντιμεταστάσεως· ὅπου γὰρ ἔστι νῦν ὕδωρ, ἐνταῦθα ἐξελθόντος
opposition; where for there is now water, here coming out
ὥσπερ ἐξ ἀγγείου πάλιν ἀὴρ ἔνεστιν, ὁτὲ δὲ τὸν αὐτὸν τόπον
as from a vessel again air is, sometimes but the same place
τοῦτον ἄλλο τι τῶν σωμάτων κατέχει· τοῦτο δὴ τῶν
this another something of the bodies contains; this indeed of the
ἐγγιγνομένων καὶ μεταβαλλόντων ἕτερον πάντων εἶναι δοκεῖ· ἐν
coming to be and changing another of all to be it seems; in
ᾧ γὰρ ἀὴρ ἔστι νῦν, ὕδωρ ἐν τούτῳ πρότερον ἦν, ὥστε δῆλον
which for air is now, water in this before was, so that it is clear
ὡς ἦν ὁ τόπος τι καὶ ἡ χώρα ἕτερον ἀμφοῖν, εἰς ἣν καὶ
that it was the place something and the land other both, into which and
ἐξ ἧς μετέβαλον.
out of which I changed.

ἔτι δὲ αἱ φοραὶ τῶν φυσικῶν σωμάτων καὶ ἁπλῶν, οἷον πυρὸς
yet and the motions of the physical bodies and simple, such as fire
καὶ γῆς καὶ τῶν τοιούτων, οὐ μόνον δηλοῦσιν ὅτι ἐστί τι
and earth and of the such, not only show that there is something

ὁ τόπος, ἀλλ' ὅτι καὶ ἔχει τινὰ δύναμιν. φέρεται γὰρ ἕκαστον
the place, but that also has some force. It is carried for each
εἰς τὸν αὐτοῦ τόπον μὴ κωλυόμενον, τὸ μὲν ἄνω τὸ δὲ κάτω·
into its own place not hindered, the one above the other below.
ταῦτα δ' ἐστὶ τόπου μέρη καὶ εἴδη, τό τε ἄνω καὶ τὸ
These indeed are parts of place and aspects, and kinds, that both above and the
κάτω καὶ αἱ λοιπαὶ τῶν ἓξ διαστάσεων. ἔστι δὲ τὰ τοιαῦτα οὐ
below and the remaining of the six dimensions. There are and the such not
μόνον πρὸς ἡμᾶς, τὸ ἄνω καὶ κάτω καὶ δεξιὸν καὶ ἀριστερόν·
only in relation to us, the above and below and right and left.
ἡμῖν μὲν γὰρ οὐκ ἀεὶ τὸ αὐτό, ἀλλὰ κατὰ τὴν θέσιν,
to us on the one hand for not always the same, but according to the position,
ὅπως ἂν στραφῶμεν, γίγνεται (διὸ καὶ ταὐτὸ
in order to (subjunctive particle) we may turn, it happens (therefore and the same
πολλάκις δεξιὸν καὶ ἀριστερὸν καὶ ἄνω καὶ κάτω καὶ πρόσθεν
many times right and left and up and down and before
καὶ ὄπισθεν), ἐν δὲ τῇ φύσει διώρισται χωρὶς ἕκαστον. οὐ
and behind), in but the nature is distinguished apart each. not
γὰρ ὅ τι ἔτυχέν ἐστι τὸ ἄνω, ἀλλ' ὅπου φέρεται τὸ
for what anything happens is the up, but wherever it is carried the
πῦρ καὶ τὸ κοῦφον· ὁμοίως δὲ καὶ τὸ κάτω οὐχ ὅ τι
fire and the light thing: similarly but and the down not what anything
ἔτυχεν, ἀλλ' ὅπου τὰ ἔχοντα βάρος καὶ τὰ γεηρά, ὡς οὐ τῇ
happens, but wherever the things having weight and the heavy, as not the
θέσει διαφέροντα μόνον ἀλλὰ καὶ τῇ δυνάμει. δηλοῖ δὲ καὶ τὰ
position differing only but also the power. it indicates but and the
μαθηματικά· οὐκ ὄντα γὰρ ἐν τόπῳ ὅμως κατὰ τὴν θέσιν τὴν
mathematics; not being for in place however according to the position the

πρὸς ἡμᾶς ἔχει δεξιὰ καὶ ἀριστερὰ ὡς τὰ μόνον λεγόμενα διὰ
towards us has right and left as the only said through
θέσιν, οὐκ ἔχοντα φύσει τούτων ἕκαστον. ἔτι οἱ τὸ κενὸν
position, not having by nature of these each. further those the void
φάσκοντες εἶναι τόπον λέγουσιν· τὸ γὰρ κενὸν τόπος ἂν εἴη
claiming to be a place they say; the for void place should be
ἐστερημένος σώματος.
deprived of body.

ὅτι μὲν οὖν ἐστί τι ὁ τόπος παρὰ τὰ σώματα, καὶ
that on the one hand therefore is something the place by the bodies, and
πᾶν σῶμα αἰσθητὸν ἐν τόπῳ, διὰ τούτων ἄν τις ὑπολάβοι·
everything body perceptible in place, through these one someone might suppose;
δόξειε δ' ἂν καὶ Ἡσίοδος ὀρθῶς λέγειν ποιήσας πρῶτον τὸ
it would seem but if also Hesiod correctly to say having created first the
χάος. λέγει γοῦν
chaos. he says at least

"πάντων μὲν πρώτιστα χάος γένετ', αὐτὰρ ἔπειτα
of all things on the one hand first and foremost chaos came to be, but afterwards
γαῖ' εὐρύστερνος," ὡς δέον πρῶτον ὑπάρξαι χώραν τοῖς
earth broad-based," as it is necessary first to exist a realm for those
οὖσι, διὰ τὸ νομίζειν, ὥσπερ οἱ πολλοί, πάντα εἶναί που καὶ
that are, due to the belief, just as the many, all to be somewhere and
ἐν τόπῳ. εἰ δ' ἐστὶ τοιοῦτο, θαυμαστή τις ἂν εἴη ἡ τοῦ
in place. if but there is such a thing, wonderful someone might be the of the
τόπου δύναμις καὶ προτέρα πάντων· οὗ γὰρ ἄνευ τῶν ἄλλων
place power and prior to all; of which for without the of others

οὐδὲν ἔστιν, ἐκεῖνο δ' [209a] ἄνευ τῶν ἄλλων, ἀνάγκη πρῶτον

nothing exists, that but [209a] without the others, necessity first

εἶναι· οὐ γὰρ ἀπόλλυται ὁ τόπος τῶν ἐν αὐτῷ φθειρομένων.

to be. not for is destroyed the place of the in it being destroyed.

οὐ μὴν ἀλλ' ἔχει γε ἀπορίαν, εἰ ἔστι, τί ἐστι, πότερον

not however but it has indeed a perplexity, if it exists, what it is, whether

ὄγκος τις σώματος ἤ τις ἑτέρα φύσις· ζητητέον γὰρ τὸ γένος

a mass some body or some other nature. to be sought for the kind

αὐτοῦ πρῶτον.

of it first.

διαστήματα μὲν οὖν ἔχει τρία, μῆκος καὶ πλάτος καὶ

Distances on the one hand therefore it has three, length and width and

βάθος, οἷς ὁρίζεται σῶμα πᾶν. ἀδύνατον δὲ σῶμα εἶναι τὸν

depth, by which it is defined body all. impossible but body to be the

τόπον· ἐν ταὐτῷ γὰρ ἂν εἴη δύο σώματα.

place. in the same for would be two bodies.

ἔτι εἴπερ ἔστι σώματος τόπος καὶ χώρα, δῆλον ὅτι καὶ

still if indeed there is of body a place and space, it is evident that and

ἐπιφανείας καὶ τῶν λοιπῶν περάτων· ὁ γὰρ αὐτὸς ἁρμόσει

of appearances and of the remaining boundaries. the for itself will fit

λόγος· ὅπου γὰρ ἦν πρότερον τὰ τοῦ ὕδατος ἐπίπεδα, ἔσται

the reason. wherever for it was previously the of the water levels, it will be

πάλιν τὰ τοῦ ἀέρος. ἀλλὰ μὴν οὐδεμίαν διαφορὰν ἔχομεν στιγμῆς

again the of the air. but indeed no difference we have of moment

καὶ τόπου στιγμῆς, ὥστ' εἰ μηδὲ ταύτης ἕτερόν ἐστιν ὁ τόπος,
and of place of moment, so that if not even of this another is the place,
οὐδὲ τῶν ἄλλων οὐδενός, οὐδ' ἐστί τι παρ' ἕκαστον τούτων
nor of the others of no one, nor is anything beside each of these
ὁ τόπος.
the place.

τί γὰρ ἄν ποτε καὶ θείημεν εἶναι τὸν τόπον; οὔτε γὰρ
what for would ever and we might define to be the place? neither for
στοιχεῖον οὔτ' ἐκ στοιχείων οἷόν τε εἶναι τοιαύτην ἔχοντα
a basic element nor from elements such as and to be such a having
φύσιν, οὔτε τῶν σω ματικῶν οὔτε τῶν ἀσωμάτων· μέγεθος
nature, nor of the so with bodies nor of the incorporeal beings. Size
μὲν γὰρ ἔχει, σῶμα δ' οὐδέν· ἔστι δὲ τὰ μὲν τῶν αἰσθητῶν
indeed for it has, body but nothing; there is and the indeed of the sensible
στοιχεῖα σώματα, ἐκ δὲ τῶν νοητῶν οὐδὲν γίγνεται μέγεθος.
elements, bodies, from and of the intelligible nothing comes to be size.

ἔτι δὲ καὶ τίνος ἄν τις θείη τοῖς οὖσιν αἴτιον εἶναι
Moreover, and even of what if someone would consider the existents cause to be
τὸν τόπον; οὐδεμία γὰρ αὐτῷ ὑπάρχει αἰτία τῶν τεττάρων· οὔτε
the place? No for to it exists cause of the four. Nor
γὰρ ὡς ὕλη τῶν ὄντων (οὐδὲν γὰρ ἐξ αὐτοῦ συνέστηκεν) οὔτε
for as matter of the beings (nothing for from it has arisen) nor
ὡς εἶδος καὶ λόγος τῶν πραγμάτων οὔθ' ὡς τέλος, οὔτε κινεῖ τὰ
as form and reason of the things nor as goal, nor moves the
ὄντα.
beings.

ἔτι δὲ καὶ αὐτὸς εἰ ἔστι τι τῶν ὄντων, ποὺ ἔσται. ἡ γὰρ
yet and also he if exists something of the beings, where it will be. the for
Ζήνωνος ἀπορία ζητεῖ τινὰ λόγον· εἰ γὰρ πᾶν τὸ ὂν ἐν τόπῳ,
Zeno's puzzle seeks some reasoning. if for everything the being in place,
δῆλον ὅτι καὶ τοῦ τόπου τόπος ἔσται, καὶ τοῦτο εἰς
it is clear that also of the place will be a place, will be, and this into
ἄπειρον.
infinity.

ἔτι ὥσπερ ἅπαν σῶμα ἐν τόπῳ, οὕτω καὶ ἐν τόπῳ ἅπαντι σῶμα·
yet just as every body in place, so also in place all body.
πῶς οὖν ἐροῦμεν περὶ τῶν αὐξανομένων; ἀνάγκη γὰρ ἐκ τούτων
how then shall we say about the increasing? necessity for from these
συναύξεσθαι αὐτοῖς τὸν τόπον, εἰ μήτ' ἐλάττων μήτε μείζων ὁ
to increase to them the place, if neither less nor greater the
τόπος ἑκάστου.
place of each.

διὰ μὲν οὖν τούτων οὐ μόνον τί ἐστιν, ἀλλὰ καὶ εἰ ἔστιν,
through indeed therefore these not only what it is, but also if it is,
ἀπορεῖν ἀναγκαῖον.
to be at a loss necessary.

Κεφάλαιο 2

Chapter 2

Ἐπεὶ δὲ τὸ μὲν καθ' αὐτὸ τὸ δὲ κατ' ἄλλο λέγεται, καὶ
Since but the one according to itself the other according to another is said, and

τόπος ὁ μὲν κοινός, ἐν ᾧ ἅπαντα τὰ σώματά ἐστιν, ὁ δ’

place the one common, in which all the bodies are, the other

ἴδιος, ἐν ᾧ πρώτῳ (λέγω δὲ οἷον σὺ νῦν ἐν τῷ οὐρανῷ ὅτι

individual, in which first I say but as you now in the sky that

ἐν τῷ ἀέρι οὗτος δ’ ἐν τῷ οὐρανῷ, καὶ ἐν τῷ ἀέρι δὲ ὅτι ἐν τῇ

in the air this and in the sky, and in the air but that in the

γῇ, ὁμοίως δὲ καὶ ἐν ταύτῃ ὅτι ἐν τῷδε τῷ τόπῳ, [209b] ὃς

earth, similarly and also in this that in this the place, [209b] which

περιέχει οὐδὲν πλέον ἢ σέ), εἰ δή ἐστιν ὁ τόπος τὸ πρῶτον

contains nothing more than you), if indeed it is the place the first

περιέχον ἕκαστον τῶν σωμάτων, πέρας τι ἂν εἴη, ὥστε

containing each of the of bodies, limit what if it may be, so that

δόξειεν ἂν τὸ εἶδος καὶ ἡ μορφὴ ἑκάστου ὁ τόπος εἶναι, ᾧ

it might seem if the form and the shape of each the place to be, to which

ὁρίζεται τὸ μέγεθος καὶ ἡ ὕλη ἡ τοῦ μεγέθους· τοῦτο γὰρ

is defined the magnitude and the matter the of the magnitude; this for

ἑκάστου πέρας. οὕτω μὲν οὖν σκοποῦσιν ὁ τόπος τὸ ἑκάστου

of each limit. Thus indeed therefore they aim the place the of each

εἶδός ἐστιν· ᾗ δὲ δοκεῖ ὁ τόπος εἶναι τὸ διάστημα τοῦ

form is; in which and it seems the place to be the distance of the

μεγέθους, ἡ ὕλη· τοῦτο γὰρ ἕτερον τοῦ μεγέθους, τοῦτο δ’ ἐστὶ

magnitude, the matter; this for another of the magnitude, this but is

τὸ περιεχόμενον ὑπὸ τοῦ εἴδους καὶ ὡρισμένον, οἷον ὑπὸ ἐπιπέδου

the content under the form and defined, such as under plane

καὶ πέρατος, ἔστι δὲ τοιοῦτον ἡ ὕλη καὶ τὸ ἀόριστον· ὅταν γὰρ

and finite, it is but such the matter and the indefinite; when for

ἀφαιρεθῇ τὸ πέρας καὶ τὰ πάθη τῆς σφαίρας, λείπεται οὐδὲν

is taken away the limit and the qualities of the sphere, nothing remains nothing

παρὰ τὴν ὕλην.

besides the matter.

διὸ καὶ Πλάτων τὴν ὕλην καὶ τὴν χώραν ταὐτό φησιν εἶναι ἐν

therefore and Plato the matter and the place the same he says to be in

τῷ Τιμαίῳ· τὸ γὰρ μεταληπτικὸν καὶ τὴν χώραν ἓν καὶ ταὐτόν.

the Timaeus. the for that which is received and the place one and the same.

ἄλλον δὲ τρόπον ἐκεῖ τε λέγων τὸ μεταληπτικὸν καὶ ἐν τοῖς

another but way there also saying the that which is received and in the

λεγομένοις ἀγράφοις δόγμασιν, ὅμως τὸν τόπον καὶ τὴν χώραν τὸ

said unwritten doctrines, however the place and the place the

αὐτὸ ἀπεφήνατο. λέγουσι μὲν γὰρ πάντες εἶναί τι τὸν τόπον,

same he declared. they say indeed for all to be something the place,

τί δ' ἐστίν, οὗτος μόνος ἐπεχείρησεν εἰπεῖν.

what then is it, this alone he attempted to say.

εἰκότως δ' ἐκ τούτων σκοπουμένοις δόξειεν ἂν εἶναι χαλεπὸν

rightly but from these who are aiming it would seem would to be difficult

γνωρίσαι τί ἐστιν ὁ τόπος, εἴπερ τούτων ὁποτερονοῦν ἐστίν,

to know what is the place, if any of these whichever it is,

εἴτε ἡ ὕλη εἴτε τὸ εἶδος· ἄλλως τε γὰρ τὴν ἀκροτάτην ἔχει

or the matter whether the form. otherwise and for the highest has

θέαν, καὶ χω ρὶς ἀλλήλων οὐ ῥᾴδιον γνωρίζειν.

sight, and in relation to one another not easy to know.

ἀλλὰ μὴν ὅτι γε ἀδύνα τον ὁποτερονοῦν τούτων εἶναι τὸν

but indeed that at least impossible the either of these to be the

τόπον, οὐ χαλεπὸν ἰδεῖν. τὸ μὲν γὰρ εἶδος καὶ ἡ ὕλη

place, not difficult to see. the on one hand for form and the matter

οὐ χωρίζεται τοῦ πράγματος, τὸν δὲ τόπον ἐνδέχεται· ἐν

not is separated from the thing, the but place may be; in

ᾧ γὰρ ἀὴρ ἦν, ἐν τούτῳ πάλιν ὕδωρ, ὥσπερ ἔφαμεν, γίγνεται,

which for air was, in this again water, as we said, becomes,

ἀντιμεθισταμένων ἀλλήλοις τοῦ τε ὕδατος καὶ τοῦ ἀέρος,

opposing to each other of the and water and of the air,

καὶ τῶν ἄλλων σωμάτων ὁμοίως, ὥστε οὔτε μόριον οὔθ' ἕξις

and of the others bodies similarly, so that neither part nor state

ἀλλὰ χωριστὸς ὁ τόπος ἑκάστου ἐστί. καὶ γὰρ δοκεῖ τοιοῦτό

but separate the place of each is. and for it seems such

τι εἶναι ὁ τόπος οἷον τὸ ἀγγεῖον (ἔστι γὰρ τὸ ἀγγεῖον τόπος

a thing to be the place as the container (it is for the container place

μεταφορητός)· τὸ δ' ἀγγεῖον οὐδὲν τοῦ πράγματός ἐστιν.

transferable) the but vessel nothing of the thing is.

ᾗ μὲν οὖν χωριστὸς [ἐστι] τοῦ πράγματος, ταύτῃ μὲν

in which indeed therefore separate [is] of the thing, in this indeed

οὐκ ἔστι τὸ εἶδος· ᾗ δὲ περιέχει, ταύτῃ δ' ἕτερος τῆς ὕλης.

not is the form· in which but contains, in this and another of the matter.

δοκεῖ δὲ ἀεὶ τὸ ὄν που αὐτό τε εἶναί τι καὶ ἕτερόν

it seems and always the being somewhere itself and to be something and another

τι ἐκτὸς αὐτοῦ. (Πλάτωνι μέντοι λεκτέον, εἰ δεῖ

something outside of it. (to Plato however it must be said, if it is necessary

παρεκβάντας εἰπεῖν, διὰ τί οὐκ ἐν τόπῳ τὰ εἴδη καὶ οἱ ἀριθμοί,

to diverge to say, because of why not in place the forms and the numbers,

εἴπερ τὸ μεθεκτικὸν ὁ τόπος, εἴτε τοῦ μεγάλου [210a] καὶ
if indeed the participant the place, whether of the large [210a] and
τοῦ μικροῦ ὄντος τοῦ μεθεκτικοῦ εἴτε τῆς ὕλης, ὥσπερ ἐν τῷ
of the small being of the participant or of the matter, just as in the
Τιμαίῳ γέγραφεν.)
Timaeus has written.)

ἔτι πῶς ἂν φέροιτο εἰς τὸν αὐτοῦ τόπον, εἰ ὁ τόπος ἡ ὕλη ἢ
yet how might be brought into the his place, if the place the matter or
τὸ εἶδος; ἀδύνατον γὰρ οὗ μὴ κίνησις μηδὲ τὸ ἄνω ἢ κάτω ἐστί,
the form? impossible for where not motion nor the up or down is,
τόπον εἶναι. ὥστε ζητητέος ἐν τοῖς τοιούτοις ὁ τόπος.
a place to be. therefore to be sought in those such the place.

εἰ δ' ἐν αὐτῷ ὁ τόπος (δεῖ γάρ, εἴπερ ἢ μορφὴ ἢ ὕλη),
if but in it the place (it is necessary for, if indeed or form or matter),
ἔσται ὁ τόπος ἐν τόπῳ· μεταβάλλει γὰρ ἅμα τῷ
there will be the place in a place; it changes for at the same time with
πράγματι καὶ κινεῖται καὶ τὸ εἶδος καὶ τὸ ἀόριστον, οὐκ ἀεὶ
the thing and it is moved and the form and the indefinite, not always
ἐν τῷ αὐτῷ ἀλλ' οὗπερ καὶ τὸ πρᾶγμα· ὥστε τοῦ τόπου
in the same but wheresoever and the thing; therefore of the place
ἔσται τόπος.
there will be place.

ἔτι ὅταν ἐξ ἀέρος ὕδωρ γένηται, ἀπόλωλεν ὁ τόπος· οὐ γὰρ
still when from air water becomes, has perished the place. not for

ἐν τῷ αὐτῷ τόπῳ τὸ γενόμενον σῶμα· τίς οὖν ἡ φθορά;

in the same place the that which has come to be body. who then the decay?

ἐξ ὧν μὲν τοίνυν ἀναγκαῖον εἶναί τι τὸν τόπον, καὶ

from which on the one hand therefore necessary to be something the place, and

πάλιν ἐξ ὧν ἀπορήσειεν ἄν τις αὐτοῦ περὶ τῆς οὐσίας,

again from which it would be at a loss if someone of it about the essence,

εἴρηται.

it has been stated.

Κεφάλαιο 3

Chapter 3

Μετὰ δὲ ταῦτα ληπτέον ποσαχῶς ἄλλο ἐν ἄλλῳ λέγεται.

After that these things it must be taken in many ways another in another is said.

ἕνα μὲν δὴ τρόπον ὡς ὁ δάκτυλος ἐν τῇ χειρὶ καὶ

one on the one hand indeed manner as the finger in the hand and

ὅλως τὸ μέρος ἐν τῷ ὅλῳ.

altogether the part in the whole.

ἄλλον δὲ ὡς τὸ ὅλον ἐν τοῖς μέρεσιν· οὐ γάρ ἐστι παρὰ τὰ μέρη

another but as the whole in the parts. not for is beside the parts

τὸ ὅλον.

the whole.

ἄλλον δὲ τρόπον ὡς ὁ ἄνθρωπος ἐν ζῴῳ καὶ ὅλως εἶδος ἐν

another but way as the man in animal and altogether species in

γένει.

genus.

ἄλλον δὲ ὡς τὸ γένος ἐν τῷ εἴδει καὶ ὅλως τὸ μέρος τοῦ
another but as the kind in the species and altogether the part of the
εἴδους ἐν τῷ λόγῳ.
species in the discourse.

ἔτι ὡς ἡ ὑγίεια ἐν θερμοῖς καὶ ψυχροῖς καὶ ὅλως τὸ εἶδος ἐν τῇ
still as the health in warm and cold and generally the nature in the
ὕλῃ.
body.

ἔτι ὡς ἐν βασιλεῖ τὰ τῶν Ἑλλήνων καὶ ὅλως ἐν τῷ πρώτῳ
still as in a king the of the Greeks and generally in the first
κινητικῷ.
cause.

ἔτι ὡς ἐν τῷ ἀγαθῷ καὶ ὅλως ἐν τῷ τέλει· τοῦτο δ' ἐστὶ τὸ
still as in the good and generally in the end; this but is the
οὗ ἕνεκα.
of which reason.

πάντων δὲ κυριώτατον τὸ ὡς ἐν ἀγγείῳ καὶ ὅλως ἐν τόπῳ.
of all and the most important the as in a vessel and generally in place.

ἀπορήσειε δ' ἄν τις, ἆρα καὶ αὐτό τι ἐν ἑαυτῷ
he would be perplexed but if anyone, whether and itself something in itself
ἐνδέχεται εἶναι, ἢ οὐδέν, ἀλλὰ πᾶν ἢ οὐδαμοῦ ἢ ἐν ἄλλῳ.
seems to be, or nothing, but everything or nowhere or in another.
διχῶς δὲ τοῦτ' ἔστιν, ἤτοι καθ' αὑτὸ ἢ καθ' ἕτερον.
in two ways and this is, either by itself or by another.

ὅταν μὲν γὰρ ᾖ μόρια τοῦ ὅλου τὸ ἐν ᾧ καὶ τὸ ἐν

when on the one hand for it is parts of the whole the in which also the in

τούτῳ, λεχθήσεται τὸ ὅλον ἐν αὑτῷ· λέγεται γὰρ καὶ κατὰ τὰ

this, it will be said the whole in itself; it is said for also according to the

μέρη, οἷον λευκὸς ὅτι ἡ ἐπιφάνεια λευκή, καὶ ἐπιστήμων ὅτι τὸ

parts, such as white that the surface is white, and knowledgeable that the

λογιστικόν. ὁ μὲν οὖν ἀμφορεὺς οὐκ ἔσται ἐν αὑτῷ, οὐδ' ὁ

rational. the indeed therefore vessel not will be in itself, nor the

οἶνος· ὁ δὲ τοῦ οἴνου ἀμφορεὺς ἔσται· ὅ τε γὰρ καὶ ἐν ᾧ,

wine; the but of the wine vessel will be; which and for also in which,

ἀμφότερα τοῦ αὐτοῦ μόρια.

both of the same parts.

οὕτω μὲν οὖν ἐνδέχεται αὐτό τι ἐν αὑτῷ εἶναι, πρώτως

thus indeed therefore it is possible this something in itself to be, first

δ' οὐκ ἐνδέχεται.

but not it is possible.

οἷον τὸ λευκὸν ἐν σώματι (ἡ [210b] ἐπιφάνεια γὰρ ἐν σώματι), ἡ

such as the white in body the [210b] appearance for in body), the

δ' ἐπιστήμη ἐν ψυχῇ· κατὰ ταῦτα δὲ αἱ προσηγορίαι μέρη

and science in soul· according to these but the designations parts

ὄντα, ὥς γε ἐν ἀνθρώπῳ (ὁ δὲ ἀμφορεὺς καὶ ὁ οἶνος χωρὶς

being, as indeed in man the but vessel and the wine apart

μὲν ὄντα οὐ μέρη, ἅμα δέ· διὸ ὅταν ᾖ μέρη, ἔσται

indeed being not parts, together but· therefore when it is parts, it will be

αὐτὸ ἐν αὑτῷ)· οἷον τὸ λευκὸν ἐν ἀνθρώπῳ ὅτι ἐν σώματι, καὶ

it in itself)· such as the white in man that in body, and

ἐν τούτῳ ὅτι ἐν ἐπιφανείᾳ· ἐν δὲ ταύτῃ οὐκέτι κατ' ἄλλο.

in this that in appearance· in but this no longer according to another.

καὶ ἕτερά γε τῷ εἴδει ταῦτα, καὶ ἄλλην φύσιν ἔχει ἑκάτερον

and different indeed to the form these, and another nature has each

καὶ δύναμιν, ἥ τ' ἐπιφάνεια καὶ τὸ λευκόν.

and force, the the appearance and the white.

οὔτε δὴ ἐπακτικῶς σκοποῦσιν οὐδὲν ὁρῶμεν ἐν ἑαυτῷ

nor indeed in an expressive manner they are aiming nothing we see in themselves

κατ' οὐδένα τῶν διορισμῶν, τῷ τε λόγῳ δῆλον ὅτι ἀδύνατον·

according to no one of the distinctions, to the and reason clear that it is impossible;

δεήσει γὰρ ἀμφότερα ἑκάτερον ὑπάρ χειν, οἷον τὸν

it will require for both each to exist as things, such as the

ἀμφορέα ἀγγεῖόν τε καὶ οἶνον εἶναι καὶ τὸν οἶνον οἰνόν τε καὶ

amphora container and and wine to be and the wine the wine and and

ἀμφορέα, εἴπερ ἐνδέχεται αὐτό τι ἐν αὑτῷ εἶναι. ὥστ' εἰ

amphora, if it is possible it something in itself to exist. so that if

ὅτι μάλιστα ἐν ἀλλήλοις εἶεν, ὁ μὲν ἀμφορεὺς δέξεται

that particularly in each other, they would be, the one amphora will accept

τὸν οἶνον οὐχ ᾗ αὐτὸς οἶνος ἀλλ' ᾗ ἐκεῖνος, ὁ δ'

the wine not in which itself it is wine but in which that one, the and

οἶνος ἐνέσται ἐν τῷ ἀμφορεῖ οὐχ ᾗ αὐτὸς ἀμφορεὺς ἀλλ' ᾗ

wine will be in the amphora not in which itself amphora but in which

ἐκεῖνος. κατὰ μὲν οὖν τὸ εἶναι ὅτι ἕτερον, δῆλον·

that one. according to on the one hand therefore the being that other, evident:

ἄλλος γὰρ ὁ λόγος τοῦ ἐν ᾧ καὶ τοῦ ἐν τούτῳ. ἀλλὰ μὴν

another for the reason of the in which and of the in this. but indeed

οὐδὲ κατὰ συμβεβηκὸς ἐνδέχεται· ἅμα γὰρ δύο ἐν
nor according to the contingent is possible: at the same time for two in
ταὐτῷ ἔσται· αὐτός τε γὰρ ἐν αὑτῷ ὁ ἀμφορεὺς ἔσται, εἰ οὗ
the same will be: he also for in itself the vessel will be, if of which
ἡ φύσις δεκτική, τοῦτ' ἐνδέχεται ἐν αὑτῷ εἶναι, καὶ ἔτι
the nature is receptive, this is possible in itself to be, and furthermore
ἐκεῖνο οὗ δεκτικόν, οἷον, εἰ οἴνου, ὁ οἶνος. ὅτι μὲν
that of which is receptive, such as, if of wine, the wine. that on the one hand
οὖν ἀδύνατον ἐν αὑτῷ τι εἶναι πρώτως, δῆλον· ὃ δὲ
therefore impossible in itself something to be first, evident: which but
Ζήνων ἠπόρει, ὅτι εἰ ὁ τόπος ἐστί τι, ἔν τινι ἔσται,
Zeno was perplexed, that if the place is something, in some it will be,
λύειν οὐ χαλεπόν· οὐδὲ γὰρ κωλύει ἐν ἄλλῳ εἶναι τὸν πρῶτον
to loosen not difficult. nor for prohibits in another to be the first
τόπον, μὴ μέντοι ὡς ἐν τόπῳ ἐκείνῳ, ἀλλ' ὥσπερ ἡ μὲν ὑγίεια
place, not however as in the place that, but just as the indeed health
ἐν τοῖς θερμοῖς ὡς ἕξις, τὸ δὲ θερμὸν ἐν σώματι ὡς πάθος.
in the warm as condition, the but heat in the body as a state.
ὥστε οὐκ ἀνάγκη εἰς ἄπειρον ἰέναι.
Therefore not necessary to the infinite to go.

ἐκεῖνο δὲ φανερόν, ὅτι ἐπεὶ οὐδὲν τὸ ἀγγεῖον τοῦ ἐν αὐτῷ (ἕτερον
that but evident, that since nothing the message of the in it (another
γὰρ τὸ πρώτως ὅ τε καὶ ἐν ᾧ), οὐκ ἂν εἴη οὔτε ἡ ὕλη
for the first which and also in which), not would be nor the matter
οὔτε τὸ εἶδος ὁ τόπος, ἀλλ' ἕτερον. ἐκείνου γάρ τι ταῦτα
nor the form the place, but another. of that for something these

τοῦ ἐνόντος, καὶ ἡ ὕλη καὶ ἡ μορφή. ταῦτα μὲν οὖν ἔστω

of existing, and the matter and the form. these indeed therefore let it be

διηπορημένα.

perplexed.

Κεφάλαιο 4

Chapter 4

Τί δέ ποτ' ἐστὶν ὁ τόπος, ᾧδ' ἂν γένοιτο φανερόν.

What then indeed is the place, in this way would become evident.

λάβωμεν δὲ περὶ αὐτοῦ ὅσα δοκεῖ ἀληθῶς καθ' αὐτὸ

let us take but concerning him as much as seems truly according to itself

ὑπάρχειν αὐτῷ.

to exist to it.

ἀξιοῦμεν δὴ τὸν τόπον εἶναι

we require indeed the place to be

1. πρῶτον μὲν [211a] περιέχον ἐκεῖνο οὗ τόπος ἐστί,

1. first indeed [211a] containing that of which place is,

2. καὶ μηδὲν τοῦ πράγματος, ἔτι τὸν πρῶτον μήτ' ἐλάττω μήτε

2. and nothing of the thing, still the first neither less nor

μείζω,

greater,

3. ἔτι ἀπολείπεσθαι ἑκάστου καὶ χωριστὸν εἶναι,

3. still to be left of each and separate to be,

4. πρὸς δὲ τούτοις πάντα τόπον ἔχειν τὸ ἄνω καὶ κάτω, καὶ

4. towards but these all place to have the above and below, and

φέρεσθαι φύσει καὶ μένειν ἐν τοῖς οἰκείοις τόποις ἕκαστον τῶν

to be carried by nature and to remain in the proper places each of the

σωμάτων, τοῦτο δὲ ποιεῖν ἢ ἄνω ἢ κάτω.

bodies, this but to make or above or below.

ὑποκειμένων δὲ τούτων τὰ λοιπὰ θεωρητέον. δεῖ δὲ

of the subjects but of these the remaining must be considered. it is necessary but

πειρᾶσθαι τὴν σκέψιν οὕτω ποιεῖσθαι ὅπως τὸ τί ἐστιν

to attempt the examination in this way to be done so that the what it is

ἀποδοθήσεται, ὥστε τά τε ἀπορούμενα λύεσθαι, καὶ

will be assigned, so that the and things that are problematic may be resolved, and

τὰ δοκοῦντα ὑπάρχειν τῷ τόπῳ ὑπάρχοντα ἔσται, καὶ ἔτι τὸ

the things that seem to exist in the place existing will be, and yet the

τῆς δυσκολίας αἴτιον καὶ τῶν περὶ αὐτὸν ἀπορημά των ἔσται

of the difficulty cause and of the concerning it problems of the will be

φανερόν· οὕτω γὰρ ἂν κάλλιστα δεικνύοιτο ἕκαστον.

evident. thus for if most excellently it would be demonstrated each one.

πρῶτον μὲν οὖν δεῖ κατανοῆσαι ὅτι οὐκ ἂν

first on the one hand therefore it is necessary to understand that not if

ἐζητεῖτο ὁ τόπος, εἰ μὴ κίνησις ἦν ἡ κατὰ τόπον· διὰ

it would be sought the place, if not motion was the according to place. for

γὰρ τοῦτο καὶ τὸν οὐρανὸν μάλιστ' οἰόμεθα ἐν τόπῳ, ὅτι ἀεὶ

indeed this and the heaven especially we believe in place, that always

ἐν κινήσει. ταύτης δὲ τὸ μὲν φορά, τὸ δὲ αὔξησις καὶ
in motion. of this and the on the one hand movement, the but increase and
φθίσις· καὶ γὰρ ἐν τῇ αὐξήσει καὶ φθίσει μεταβάλλει, καὶ ὃ
decrease; and for in the increase and decrease changes, and that which
πρότερον ἦν ἐνταῦθα, πάλιν μεθέστηκεν εἰς ἔλαττον ἢ μεῖζον.
previously was here, again has changed into lesser or greater.

ἔστι δὲ κινούμενον τὸ μὲν καθ' αὑτὸ ἐνεργείᾳ, τὸ δὲ
it is and moving the on the one hand according to itself activity, the but
κατὰ συμβεβηκός· κατὰ συμβεβηκὸς δὲ τὸ μὲν
according to the accidental; according to the accidental and the on the one hand
ἐνδεχόμενον κινεῖσθαι καθ' αὑτό, οἷον τὰ μόρια τοῦ
possible to be moved according to itself, such as the particles of the
σώματος καὶ ὁ ἐν τῷ πλοίῳ ἧλος, τὰ δ' οὐκ ἐνδεχόμενα ἀλλ'
body and the in the ship pin, the but not possible but
ἀεὶ κατὰ συμβεβηκός, οἷον ἡ λευκότης καὶ ἡ ἐπιστήμη·
always according to the accidental, such as the whiteness and the science;
ταῦτα γὰρ οὕτω μεταβέβληκε τὸν τόπον, ὅτι ἐν ᾧ ὑπάρχουσι
these for thus has changed the place, that in which exist
μεταβάλλει.
changes.

ἐπεὶ δὲ λέγομεν εἶναι ὡς ἐν τόπῳ ἐν τῷ οὐρανῷ, διότι ἐν τῷ ἀέρι
since but we say to be as in place in the sky, because in the air
οὗτος δὲ ἐν τῷ οὐρανῷ· καὶ ἐν τῷ ἀέρι δὲ οὐκ ἐν παντί, ἀλλὰ
this but in the sky. and in the air but not in everything, but
διὰ τὸ ἔσχατον αὐτοῦ καὶ περιέχον ἐν τῷ ἀέρι φαμὲν εἶναι (εἰ
through the ultimate of it and surrounding in the air we say to be (if

γὰρ πᾶς ὁ ἀὴρ τόπος, οὐκ ἂν ἴσος εἴη ἑκάστου ὁ τόπος καὶ
for every the air place, not would equal be of each the place and
ἕκαστον, δοκεῖ δέ γε ἴσος εἶναι, τοιοῦτος δ' ὁ πρῶτος ἐν
each, it seems but at least equal to be, such indeed the first in
ᾧ ἐστιν)·
which it is).

ὅταν μὲν οὖν μὴ διῃρημένον ᾖ τὸ περιέχον ἀλλὰ συνεχές, οὐχ
whenever indeed therefore not divided be the containing but continuous, not
ὡς ἐν τόπῳ λέγεται εἶναι ἐν ἐκείνῳ, ἀλλ' ὡς μέρος ἐν ὅλῳ· ὅταν
as in place it is said to be in that, but as a part in all. whenever
δὲ διῃρημένον ᾖ καὶ ἁπτόμενον, ἐν πρώτῳ ἐστὶ τῷ ἐσχάτῳ
but divided may be and touching, in the first there is the last
τοῦ περιέχοντος, ὃ οὔτε ἐστὶ μέρος τοῦ ἐν αὐτῷ οὔτε μεῖζον
of the containing, which neither is a part of the in it nor greater
τοῦ διαστήματος ἀλλ' ἴσον· ἐν γὰρ τῷ αὐτῷ τὰ ἔσχατα τῶν
of the interval but equal. in for the same the farthest of the
ἁπτομένων.
touching.

καὶ συνεχὲς μὲν ὂν οὐκ ἐν ἐκείνῳ κινεῖται ἀλλὰ μετ' ἐκείνου,
and continuous on one hand being not in that moves but with that,
διῃρημένον δὲ ἐν ἐκείνῳ· καὶ ἐάν τε κινῆται τὸ περιέχον ἐάν τε
divided but in that. and if also it moves the containing if also
μή, [211b] οὐδὲν ἧττον.
not, [211b] nothing less.

[ἔτι ὅταν μὴ διῃρημένον ᾖ, ὡς μέρος ἐν ὅλῳ λέγεται,
[Moreover whenever not divided may be, as a part in all is said,

οἷον ἐν τῷ ὀφθαλμῷ ἡ ὄψις ἢ ἐν τῷ σώματι ἡ χείρ, ὅταν δὲ

such as in the eye the sight or in the body the hand when but

διῃρημένον, οἷον ἐν τῷ κάδῳ τὸ ὕδωρ ἢ ἐν τῷ κεραμίῳ ὁ οἶνος·

divided such as in the jar the water or in the earthenware the wine

ἡ μὲν γὰρ χεὶρ μετὰ τοῦ σώματος κινεῖται, τὸ δὲ ὕδωρ ἐν τῷ

the on one hand for hand with the body moves the but water in the

κάδῳ.]

jar

ἤδη τοίνυν φανερὸν ἐκ τούτων τί ἐστιν ὁ τόπος. σχεδὸν γὰρ

already therefore evident from these what is the place. almost for

τέτταρά ἐστιν ὧν ἀνάγκη τὸν τόπον ἕν τι εἶναι· ἢ γὰρ

four is of which necessity the place one something to be. or for

μορφὴ ἢ ὕλη ἢ διάστημά τι τὸ μεταξὺ τῶν ἐσχάτων, ἢ τὰ

form or matter or interval something the between the extremes, or the

ἔσχατα εἰ μὴ ἔστι μηδὲν διάστημα παρὰ τὸ τοῦ ἐγγιγνομένου

extremes if not is nothing interval besides the of the coming into being

σώματος μέγεθος.

body size.

τούτων δ' ὅτι οὐκ ἐνδέχεται τὰ τρία εἶναι, φανερόν· ἀλλὰ

Of these but that it is not possible the three to be, evident; but

διὰ μὲν τὸ περιέχειν δοκεῖ ἡ μορφὴ εἶναι· ἐν ταὐτῷ γὰρ τὰ

through on one hand the containing seems the form to be. In the same for the

ἔσχατα τοῦ περιέχοντος καὶ τοῦ περιεχομένου. ἔστι μὲν οὖν

extremes of the containing and of the contained. It is indeed therefore

ἄμφω πέρατα, ἀλλ' οὐ τοῦ αὐτοῦ, ἀλλὰ τὸ μὲν εἶδος τοῦ

both extremes, but not of the same, but the on one hand form of the

πράγματος, ὁ δὲ τόπος τοῦ περιέχοντος σώματος.

thing, the on the other hand place of the containing body.

διὰ δὲ τὸ μεταβάλλειν πολλάκις μένοντος τοῦ περιέχοντος τὸ

Through but the changing often remaining of the containing the

περιεχόμενον καὶ διῃρημένον, οἷον ἐξ ἀγγείου ὕδωρ, τὸ μεταξὺ

content and divided, such as from a vessel water, the between

εἶναί τι δοκεῖ διάστημα, ὡς ὄν τι παρὰ τὸ σῶμα τὸ

to be something it seems a distance, as being something near the body the

μεθιστάμενον. τὸ δ' οὐκ ἔστιν, ἀλλὰ τὸ τυχὸν ἐμπίπτει σῶμα

that is changing. the and not is, but the accidental falls body

τῶν μεθισταμένων καὶ ἅπτεσθαι πεφυκότων.

of the changing and to touch of those that exist.

εἰ δ' ἦν τι [τὸ] διάστημα <καθ' αὑ>τὸ πεφυκὸς

if but there was something [the] interval according to itself having originated

<εἶναι> καὶ μένον, ἐν τῷ αὐτῷ ἄπειροι ἂν ἦσαν τόποι (μεθισταμένου

to be and remaining, in the same infinite would be places (changing

γὰρ τοῦ ὕδατος καὶ τοῦ ἀέρος ταὐτὸ ποιήσει τὰ μόρια πάντα ἐν

for the water and the air the same will produce the particles all in

τῷ ὅλῳ ὅπερ ἅπαν τὸ ὕδωρ ἐν τῷ ἀγγείῳ)· ἅμα δὲ καὶ ὁ

the whole which all the water in the vessel). at the same time but and the

τόπος ἔσται μεταβάλλων· ὥστ' ἔσται τοῦ τόπου τ' ἄλλος

place will be changing. so that there will be of the place another other

τόπος, καὶ πολλοὶ τόποι ἅμα ἔσονται. οὐκ ἔστι δὲ ἄλλος

place, and many places at the same time will exist. not is but another

ὁ τόπος τοῦ μορίου, ἐν ᾧ κινεῖται, ὅταν ὅλον τὸ ἀγγεῖον

the place of the particle, in which it moves, when all the vessel

μεθίστηται, ἀλλ' ὁ αὐτός· ἐν ᾧ γὰρ ἔστιν, ἀντιμεθίσταται

is changed, but the same; in which for it is, opposes

ὁ ἀὴρ καὶ τὸ ὕδωρ ἢ τὰ μόρια τοῦ ὕδατος, ἀλλ' οὐκ ἐν ᾧ

the air and the water or the parts of the water, but not in which

γίγνονται τόπῳ, ὃς μέρος ἐστὶ τοῦ τόπου ὅς ἐστι τόπος

they become place, which part is of the place which is a place

ὅλου τοῦ οὐρανοῦ.

of the whole of the sky.

καὶ ἡ ὕλη δὲ δόξειεν ἂν εἶναι τόπος, εἴ γε ἐν ἠρεμοῦντί

and the wood but it would seem would to be a place, if indeed in tranquility

τις σκοποίη καὶ μὴ κεχωρισμένῳ ἀλλὰ συνεχεῖ. ὥσπερ γὰρ

someone wondering and not separated but continuous. just as for

εἰ ἀλλοιοῦται, ἔστι τι ὃ νῦν μὲν λευκὸν πάλαι δὲ

if it is altered, there is something which now on one hand white long ago but

μέλαν, καὶ νῦν μὲν σκληρὸν πάλαι δὲ μαλακόν (διό φαμεν

black, and now on one hand hard long ago but soft (therefore we say

εἶναί τι τὴν ὕλην), οὕτω καὶ ὁ τόπος διὰ τοιαύτης τινὸς

to be something the wood), thus and the place through such a kind

εἶναι δοκεῖ φαντασίας, πλὴν ἐκεῖνο μὲν διότι ὃ ἦν ἀήρ,

to be it seems of imagination, except that on one hand because which was air,

τοῦτο νῦν ὕδωρ, ὁ δὲ τόπος ὅτι οὗ ἦν ἀήρ, ἐνταῦθ' ἔστι

this now water, the but place that of which it was air, here there is

νῦν ὕδωρ. ἀλλ' ἡ μὲν ὕλη, [212a] ὥσπερ ἐλέχθη ἐν τοῖς

now water. but the on one hand wood, [212a] just as it was said in the

πρότερον, οὔτε χωριστὴ τοῦ πράγματος οὔτε περιέχει, ὁ δὲ

former, neither separate of the thing nor contains, the but

τόπος ἄμφω.

place both.

εἰ τοίνυν μηδὲν τῶν τριῶν ὁ τόπος ἐστίν, μήτε τὸ εἶδος μήτε

if then nothing of the three the place is, neither the form nor

ἡ ὕλη μήτε διάστημά τι ἀεὶ ὑπάρχον ἕτερον παρὰ τὸ τοῦ

the matter nor dimension anything always existing other beside the of the

πράγματος τοῦ μεθισταμένου, ἀνάγκη τὸν τόπον εἶναι τὸ λοιπὸν

thing the changing, necessity the place to be the remaining

τῶν τεττά ρων, τὸ πέρας τοῦ περιέχοντος σώματος <καθ'

of the four, parts, the boundary of the containing body <according to

ὃ συνάπτει τῷ περιεχομένῳ>. λέγω δὲ τὸ περιεχόμενον σῶμα

which connects to the contained.> I say but the contained body

τὸ κινητὸν κατὰ φοράν.

the movable according to motion.

δοκεῖ δὲ μέγα τι εἶναι καὶ χαλεπὸν ληφθῆναι ὁ τόπος διά

it seems but great something to be and difficult to be taken the place through

τε τὸ παρεμφαίνεσθαι τὴν ὕλην καὶ τὴν μορφήν, καὶ διὰ τὸ ἐν

and the appearing the matter and the form, and through the in

ἠρεμοῦντι τῷ περιέχοντι γίγνεσθαι τὴν μετάστασιν τοῦ φερομένου·

settling the containing to become the transition of the carried.

ἐνδέχεσθαι γὰρ φαί νεται εἶναι διάστημα μεταξὺ ἄλλο

it is possible for it appears to happen to be a distance between another

τι τῶν κινουμένων μεγεθῶν. συμβάλλεται δέ τι καὶ ὁ

something of the moving magnitudes. it converges but something and the

ἀὴρ δοκῶν ἀσώματος εἶναι· φαίνεται γὰρ οὐ μόνον τὰ πέρατα

air seeming incorporeal to be. it appears for not only the limits

τοῦ ἀγγείου εἶναι ὁ τόπος, ἀλλὰ καὶ τὸ μεταξὺ ὡς κενὸν
of the vessel to be the place, but also and the between as a void
<ὄν>. ἔστι δ' ὥσπερ τὸ ἀγγεῖον τόπος μεταφορητός, οὕτως
[being]. there is but just as the vessel place transportable, so
καὶ ὁ τόπος ἀγγεῖον ἀμετακί νητον. διὸ ὅταν μὲν ἐν
and the place vessel unchanging stable. therefore when on the one hand in
κινουμένῳ κινῆται καὶ μεταβάλλῃ τὸ ἐντός, οἷον ἐν ποταμῷ πλοῖον,
motion it moves and changes the interior, as if in a river a vessel,
ὡς ἀγγείῳ χρῆται μᾶλλον ἢ τόπῳ τῷ περιέχοντι. βούλεται δ'
as a vessel it is used more than a place that contains. it desires but
ἀκίνητος εἶναι ὁ τόπος· διὸ ὁ πᾶς μᾶλλον ποταμὸς τόπος,
to be at rest to be the place; therefore the whole more river is a place,
ὅτι ἀκίνητος ὁ πᾶς.
because it is at rest the whole.

ὥστε τὸ τοῦ περιέχοντος πέρας ἀκίνητον πρῶτον, τοῦτ' ἔστιν ὁ
hence the of the containing boundary is unchanging first, this is the
τόπος.
place.

καὶ διὰ τοῦτο τὸ μέσον τοῦ οὐρανοῦ καὶ τὸ ἔσχατον τὸ πρὸς
and through this the middle of the sky and the utmost the towards
ἡμᾶς τῆς κύκλῳ φορᾶς δοκεῖ εἶναι τὸ μὲν ἄνω τὸ δὲ κάτω
us of the circle motion seems to be the on one hand above the but below
μάλιστα πᾶσι κυρίως, ὅτι τὸ μὲν αἰεὶ μένει, τοῦ δὲ
especially for all mainly, because the on one hand always remains, of the but
κύκλῳ τὸ ἔσχατον ὡσαύτως ἔχον μένει. ὥστ' ἐπεὶ τὸ μὲν
circle the utmost likewise having remains. so since the on one hand

κοῦφον τὸ ἄνω φερόμενόν ἐστι φύσει, τὸ δὲ βαρὺ τὸ κάτω, τὸ

light the above carried is by nature, the but heavy the below, the

μὲν πρὸς τὸ μέσον περιέχον πέρας κάτω ἐστίν, καὶ αὐτὸ τὸ

on one hand towards the middle surrounding limit below is, and it the

μέσον, τὸ δὲ πρὸς τὸ ἔσχατον ἄνω, καὶ αὐτὸ τὸ ἔσχατον·

middle, the but towards the utmost above, and it the utmost.

καὶ διὰ τοῦτο δοκεῖ ἐπίπεδόν τι εἶναι καὶ οἷον ἀγγεῖον ὁ

and through this seems flat something to be and such as a container the

τόπος καὶ περιέχον.

place and containing.

ἔτι ἅμα τῷ πράγματι ὁ τόπος· ἅμα γὰρ τῷ

yet at the same time with the thing the place. at the same time for with the

πεπερασμένῳ τὰ πέρατα.

finite the boundaries.

Κεφάλαιο 5

Chapter 5

Ὧι μὲν οὖν σώματι ἔστι τι ἐκτὸς σῶμα περιέχον

To which indeed therefore body is something external body containing

αὐτό, τοῦτο ἔστιν ἐν τόπῳ, ᾧ δὲ μή, οὔ. διὸ κἂν ὕδωρ

it, this is in place, to which but not, no. therefore even if water

γένηται τοιοῦτο, τὰ μὲν μόρια κινήσεται αὐτοῦ (περιέχεται

were to become such, the indeed parts will move of itself (is contained

γὰρ ὑπ' ἀλλήλων), τὸ δὲ πᾶν ἔστι μὲν ὡς κινήσεται ἔστι δ' ὡς

for by each other), the but all is indeed as it moves it is and as

οὔ. ὡς μὲν γὰρ ὅλον, ἅμα τὸν τόπον οὐ μεταβάλλει,

not. as indeed for the whole, at the same time the place not changes,

κύκλῳ [212b] δὲ κινεῖται-τῶν μορίων γὰρ οὗτος ὁ τόπος-καὶ

in a circle [212b] but is moved-of parts for this the place-and

ἄνω μὲν καὶ κάτω οὔ, κύκλῳ δ' ἔνια· τὰ δὲ καὶ ἄνω καὶ

above on the one hand and below not, in a circle but some; the but and above and

κάτω, ὅσα ἔχει πύκνωσιν καὶ μάνωσιν.

below, as much as has density and quality.

ὥσπερ δ' ἐλέχθη, τὰ μέν ἐστιν ἐν τόπῳ κατὰ δύναμιν,

just as but was said, the on the one hand are in place according to power,

τὰ δὲ κατ' ἐνέργειαν. διὸ ὅταν μὲν συνεχὲς

the on the other hand according to activity. therefore when on one hand continuous

ᾖ τὸ ὁμοιομερές, κατὰ δύναμιν ἐν τόπῳ τὰ μέρη, ὅταν δὲ

is the homogeneous, according to power in place the parts, when but

χωρισθῇ μὲν ἅπτηται δ' ὥσπερ σωρός, κατ' ἐνέργειαν.

is separated on one hand touches but just as a heap, according to activity.

καὶ τὰ μὲν καθ' αὑτά (οἷον πᾶν σῶμα ἢ κατὰ φορὰν

and the on the one hand by themselves (like every body or according to motion

ἢ κατ' αὔξησιν κινητὸν καθ' αὑτό που, ὁ δ' οὐρανός,

or according to growth a mobile by itself somewhere, the but heaven,

ὥσπερ εἴρηται, οὗ που ὅλος οὐδ' ἔν τινι τόπῳ ἐστίν, εἴ

just as has been said, not anywhere entire nor in any place is, if

γε μηδὲν αὐτὸν περιέχει σῶμα· ἐφ' ᾧ δὲ κινεῖται, ταύτῃ

indeed nothing itself contains body. on which it but is moved, in this way

καὶ τόπος ἔστι τοῖς μορίοις· ἕτερον γὰρ ἑτέρου ἐχόμενον τῶν

and place is to the parts another for of another holding of the

μορίων ἐστίν)·

parts it is)

τὰ δὲ κατὰ συμβεβηκός, οἷον ἡ ψυχὴ καὶ ὁ οὐρανός·
the but according to that which has happened, such as the soul and the sky·
τὰ γὰρ μόρια ἐν τόπῳ πως πάντα· ἐπὶ τῷ κύκλῳ γὰρ περιέχει
the for parts in place somehow all· upon the circle for contains
ἄλλο ἄλλο. διὸ κινεῖται μὲν κύκλῳ τὸ ἄνω, τὸ
one another another. therefore it moves on the one hand in a circle, the upper, the
δὲ πᾶν οὔ που. τὸ γάρ που αὐτό τέ ἐστί τι, καὶ ἔτι
but all not anywhere. the for somewhere itself and is something, and still
ἄλλο τι δεῖ εἶναι παρὰ τοῦτο ἐν ᾧ, ὃ περιέχει· παρὰ δὲ
another something must be besides this in which, which contains· besides but
τὸ πᾶν καὶ ὅλον οὐδέν ἐστιν ἔξω τοῦ παντός, καὶ διὰ τοῦτο
the all and whole nothing is outside of the whole, and through this
ἐν τῷ οὐρανῷ πάντα· ὁ γὰρ οὐρανὸς τὸ πᾶν ἴσως. ἔστι δ' ὁ
in the sky all· the for sky the all perhaps. it is but the
τόπος οὐχ ὁ οὐρανός, ἀλλὰ τοῦ οὐρανοῦ τι τὸ ἔσχατον καὶ
place not the sky, but of the sky something the utmost and
ἁπτόμενον τοῦ κινητοῦ σώματος [πέρας ἠρεμοῦν]. καὶ διὰ τοῦτο
touching of the mobile body limit of rest]. and through this
ἡ μὲν γῆ ἐν τῷ ὕδατι, τοῦτο δ' ἐν τῷ ἀέρι, οὗτος δ'
the on the one hand earth in the water, this but in the air, this but
ἐν τῷ αἰθέρι, ὁ δ' αἰθὴρ ἐν τῷ οὐρανῷ, ὁ δ' οὐρανὸς οὐκέτι ἐν
in the ether, the but ether in the sky, the but sky no longer in
ἄλλῳ.
another.

φανερὸν δ' ἐκ τούτων ὅτι καὶ αἱ ἀπορίαι πᾶσαι λύοιντ' ἂν
evident but from these that also the difficulties all would be resolved if

οὕτω λεγομένου τοῦ τόπου.

thus being said of the place.

1. οὔτε γὰρ συναύξεσθαι ἀνάγκη τὸν τόπον,

1. neither for to be enlarged is necessary the place,

2. οὔτε στιγμῆς εἶναι τόπον, οὔτε δύο σώματα ἐν τῷ αὐτῷ τόπῳ,

2. nor of a moment to be place, nor two bodies in the to him place,

3. οὔτε διάστημά τι εἶναι σωματικόν (σῶμα γὰρ τὸ μεταξὺ τοῦ τόπου τὸ τυχόν, ἀλλ' οὐ διάστημα σώματος).

3. nor space anything to be corporeal (body for the between of the place the possible, but not space of a body).

4. καὶ ἔστιν ὁ τόπος καὶ πού, οὐχ ὡς ἐν τόπῳ δέ, ἀλλ' ὡς τὸ πέρας ἐν τῷ πεπερασμένῳ. οὐ γὰρ πᾶν τὸ ὂν ἐν τόπῳ, ἀλλὰ τὸ κινητὸν σῶμα.

4. and is the place and where, not as in place, but, rather as the limit in the finite. not for all the being in place, but the mobile body.

5. καὶ φέρεται δὴ εἰς τὸν αὑτοῦ τόπον ἕκαστον εὐλόγως (ὃ γὰρ ἐφεξῆς καὶ ἁπτόμενον μὴ βίᾳ, συγγενές· καὶ συμπεφυκότα μὲν ἀπαθῆ, ἁπτόμενα δὲ παθητικὰ καὶ ποιητικὰ

5. And it is brought indeed to its own place each appropriately (which for subsequently and touching not by force, related; and having been united indeed unmoved, touching but passive and creative

ἀλλήλων)· καὶ μένει δὴ φύσει πᾶν ἐν τῷ οἰκείῳ τόπῳ οὐκ
of each other); and remains indeed by nature all in the proper place not
ἀλόγως· καὶ γὰρ τὸ μέρος, τὸ δὲ ἐν [τῷ] τόπῳ ὡς διαιρετὸν μέρος
irrationally; and for the part, the but in [the] place as divisible part
πρὸς ὅλον ἐστίν, οἷον ὅταν ὕδατος κινήσῃ τις [213a] μόριον
towards the whole is, such as when of water (it) moves someone [213a] a part
ἢ ἀέρος. οὕτω δὲ καὶ ἀὴρ ἔχει πρὸς ὕδωρ· οἷον ὕλη γάρ, τὸ
or of air. In this way but also air has towards water; such as matter for, the
δὲ εἶδος, τὸ μὲν ὕδωρ ὕλη ἀέρος, ὁ δ' ἀὴρ οἷον ἐνέργειά
but form, the on the one hand water matter of air, the and air such as a potency
τις ἐκείνου· τὸ γὰρ ὕδωρ δυνάμει ἀήρ ἐστιν, ὁ δ' ἀὴρ δυνάμει
some of that; the for water in power (is) air, it is, the and air in power
ὕδωρ ἄλλον τρόπον.
water another way.

διοριστέον δὲ περὶ τούτων ὕστερον· ἀλλὰ διὰ τὸν
it is necessary to appoint but concerning these later but through the
καιρὸν ἀνάγκη μὲν εἰπεῖν, ἀσαφῶς δὲ νῦν ῥηθὲν τότ'
time necessity indeed to say, obscurely but now having been said then
ἔσται σαφέστερον. εἰ οὖν τὸ αὐτὸ [ἡ] ὕλη καὶ ἐντελέχεια
it will be clearer. if therefore the same [the] matter and actuality
(ὕδωρ γὰρ ἄμφω, ἀλλὰ τὸ μὲν δυνάμει τὸ δ' ἐντελεχείᾳ),
(water for both, but the on one hand in potentiality the and in actuality),
ἔχοι ἂν ὡς μόριόν πως πρὸς ὅλον. διὸ καὶ τούτοις
it would have could as a part somehow towards the whole. therefore and by these
ἁφὴ ἔστιν· σύμφυσις δέ, ὅταν ἄμφω ἐνεργείᾳ ἓν γένωνται. καὶ
touch is; co but, when both in actuality one come to be. and

περὶ μὲν τόπου, καὶ ὅτι ἔστι καὶ τί ἐστιν, εἴρηται.

concerning indeed of place, and that it is and what it is, has been said.

Κεφάλαιο 6

Chapter 6

Τὸν αὐτὸν δὲ τρόπον ὑποληπτέον εἶναι τοῦ φυσικοῦ θεωρῆσαι

The same but manner must be assumed to be of the natural to contemplate

καὶ περὶ κενοῦ, εἰ ἔστιν ἢ μή, καὶ πῶς ἔστι, καὶ τί ἐστιν,

and concerning emptiness, if it is or not, and how it is, and what it is,

ὥσπερ καὶ περὶ τόπου· καὶ γὰρ παραπλησίαν ἔχει τήν τε

just as and regarding place; and for a similar has the and

ἀπιστίαν καὶ τὴν πίστιν διὰ τῶν ὑπολαμβανομένων· οἷον γὰρ

distrust and the belief through the assumptions; for example, for

τόπον τινὰ καὶ ἀγγεῖον τὸ κενὸν τιθέασιν οἱ λέγοντες, δοκεῖ δὲ

a place some and vessel the empty they posit the speakers, it seems but

πλῆρες μὲν εἶναι, ὅταν ἔχῃ τὸν ὄγκον οὗ δεκτικόν

full indeed to be, whenever it possesses the mass of which it is receptive,

ἐστιν, ὅταν δὲ στερηθῇ, κενόν, ὡς τὸ αὐτὸ μὲν ὂν κενὸν

it is, whenever but it is deprived, empty, just as the same indeed being empty

καὶ πλῆρες καὶ τόπον, τὸ δ' εἶναι αὐτοῖς οὐ ταὐτὸ ὄν.

and full and a place, the yet to be to them not the same being.

ἄρξασθαι δὲ δεῖ τῆς σκέψεως λαβοῦσιν ἅ τε λέγουσιν

to begin but it is necessary of the thought to take whatever and they say

οἱ φάσκοντες εἶναι καὶ πάλιν ἃ λέγουσιν οἱ μὴ φάσκοντες, καὶ

the claiming to be and again what they say the not claiming, and

τρίτον τὰς κοινὰς περὶ αὐτῶν δόξας.

third the common about them opinions.

οἱ μὲν οὖν δεικνύναι πειρώμενοι ὅτι οὐκ ἔστιν, οὐχ ὃ
the indeed therefore to demonstrate attempting that not is, not what
βούλονται λέγειν οἱ ἄνθρωποι κενόν, τοῦτ' ἐξελέγχουσιν, ἀλλ'
they wish to say the people empty, this they refute, but
<ὃ> ἁμαρτάνοντες λέγουσιν. ὥσπερ Ἀναξαγόρας καὶ οἱ τοῦτον
what erring they say. just as Anaxagoras and the those
τὸν τρόπον ἐλέγχοντες. ἐπιδεικνύουσι γὰρ ὅτι ἐστίν τι ὁ
that manner refuting. they demonstrate for that there is something the
ἀήρ, στρεβλοῦντες τοὺς ἀσκοὺς καὶ δεικνύντες ὡς ἰσχυρὸς ὁ
air, distorting the vessels and showing that strong the
ἀήρ, καὶ ἐναπολαμβάνοντες ἐν ταῖς κλεψύδραις. οἱ δὲ ἄνθρωποι
air, and also receiving in the clepsydras. the but people
βούλονται κενὸν εἶναι διάστημα ἐν ᾧ μηδέν ἐστι σῶμα αἰσθητόν·
wish an empty to be space in which nothing is body perceptible;
οἰόμενοι δὲ τὸ ὂν ἅπαν εἶναι σῶμα φασίν, ἐν ᾧ ὅλως μηδέν
believing but the being all to be body they claim, in which totally nothing
ἐστι, τοῦτ' εἶναι κενόν, διὸ τὸ πλῆρες ἀέρος κενὸν εἶναι.
is, this to be empty, therefore the full of air empty to be.
οὔκουν τοῦτο δεῖ δεικνύναι, ὅτι ἐστί τι ὁ ἀήρ, ἀλλ'
Therefore not this must show, that it is something the air, but
ὅτι οὐκ ἔστι διάστημα ἕτερον τῶν σωμάτων, οὔτε χωριστὸν
that not is a distance different of the bodies, neither separate
οὔτε ἐνεργείᾳ ὄν, ὃ διαλαμβάνει τὸ πᾶν σῶμα ὥστε εἶναι μὴ
nor with energy being, which includes the whole body so that it is not
συνεχές, καθάπερ λέγουσιν Δημόκριτος καὶ Λεύκιππος καὶ [213b]
continuous, as they say Democritus and Leucippus and [213b]

ἕτεροι πολλοὶ τῶν φυσιολόγων, ἢ καὶ εἴ τι ἔξω τοῦ παντὸς
others many of the philosophers, or even if something outside of the whole
σώματός ἐστιν ὄντος συνεχοῦς.
body exists being continuous.

οὗτοι μὲν οὖν οὐ κατὰ θύρας πρὸς τὸ πρόβλημα ἀπαντῶσιν,
These indeed therefore not according to doors towards the problem they respond,
ἀλλ' οἱ φάσκοντες εἶναι μᾶλλον.
but those claiming to be more so.

λέγουσιν δ' ἓν μὲν ὅτι κίνησις ἡ κατὰ τόπον οὐκ ἂν
they say but one on the one hand that motion the according to place not would
εἴη (αὕτη δ' ἐστὶ φορὰ καὶ αὔξησις)· οὐ γὰρ ἂν δοκεῖν εἶναι
be this but is a carrying and increase) not for would seem to be
κίνησιν, εἰ μὴ εἴη κενόν· τὸ γὰρ πλῆρες ἀδύνατον εἶναι δέξασθαί
motion, if not be empty; the for full impossible to be to receive
τι. εἰ δὲ δέξεται καὶ ἔσται δύο ἐν ταὐτῷ, ἐνδέχοιτ'
something. if but it receives and will be two in the same, it would be possible
ἂν καὶ ὁποσαοῦν εἶναι ἅμα σώματα· τὴν γὰρ διαφοράν,
would and whatever to be at the same time bodies; the for difference,
δι' ἣν οὐκ ἂν εἴη τὸ λεχθέν, οὐκ ἔστιν εἰπεῖν. εἰ δὲ τοῦτο
through which not would be the said, not is to say. if but this
ἐνδέχεται, καὶ τὸ μικρότατον δέξεται τὸ μέγιστον· πολλὰ γὰρ
is possible, and the smallest will be received the greatest; many for
μικρὰ τὸ μέγα ἐστίν· ὥστε εἰ πολλὰ ἴσα ἐνδέχεται ἐν ταὐτῷ
small the great is; so that if many equal it is possible in the same
εἶναι, καὶ πολλὰ ἄνισα.
to be, and many unequal.

Μέλισσος μὲν οὖν καὶ δείκνυσιν ὅτι τὸ πᾶν ἀκίνητον ἐκ
Melissus indeed therefore and shows that the whole is immovable from
τούτων· εἰ γὰρ κινήσεται, ἀνάγκη εἶναι (φησί) κενόν, τὸ δὲ κενὸν
those; if for it moves, necessity to be (he says) empty, the but empty
οὐ τῶν ὄντων.
not of the beings.

ἕνα μὲν οὖν τρόπον ἐκ τούτων δεικνύουσιν ὅτι ἔστιν τι
one indeed therefore a way from these they show that there is something
κενόν,
empty,

ἄλλον δ' ὅτι φαίνεται ἔνια συνιόντα καὶ πιλούμενα, οἷον καὶ τὸν
another but that appears some coming together and being pressed, such as and the
οἶνόν φασι δέχεσθαι μετὰ τῶν ἀσκῶν τοὺς πίθους, ὡς εἰς τὰ
wine they say to receive along with the skins the jars, as into the
ἐνόντα κενὰ συνιόντος τοῦ πυκνουμένου σώματος.
existing emptiness of the coming together of the compacting body.

ἔτι δὲ καὶ ἡ αὔξησις δοκεῖ πᾶσι γίγνεσθαι διὰ κενοῦ· τὴν
still and also the increase seems to all to be becoming through of the void. the
μὲν γὰρ τροφὴν σῶμα εἶναι, δύο δὲ σώματα ἀδύνατον
on one hand for nourishment to be a body, to be, two and bodies impossible
ἅμα εἶναι. μαρτύριον δὲ καὶ τὸ περὶ τῆς τέφρας ποιοῦνται,
at the same time to be. testimony and also the concerning the ash they produce,
ἣ δέχεται ἴσον ὕδωρ ὅσον τὸ ἀγγεῖον τὸ κενόν. εἶναι δ'
which receives equal water as much as the vessel the void. to be but

ἔφασαν καὶ οἱ Πυθαγόρειοι κενόν, καὶ ἐπεισιέναι αὐτὸ τῷ
they said and the Pythagoreans void, and to enter it to the
οὐρανῷ ἐκ τοῦ ἀπείρου πνεύματος ὡς ἀναπνέοντι καὶ τὸ κενόν,
heaven from the infinite spirit as breathing and the void,
ὃ διορίζει τὰς φύσεις, ὡς ὄντος τοῦ κενοῦ χωρισμοῦ τινὸς
which separates the natures, as being the void's separation of a certain
τῶν ἐφεξῆς καὶ [τῆς] διορίσεως· καὶ τοῦτ' εἶναι πρῶτον ἐν τοῖς
of the following and [the] separation; and this is first in the
ἀριθμοῖς· τὸ γὰρ κενὸν διορίζειν τὴν φύσιν αὐτῶν.
by numbers. The for void to distinguish the nature of them.

ἐξ ὧν μὲν οὖν οἱ μέν φασιν εἶναι οἱ δ' οὔ φασι,
out of which on one hand therefore the ones say to be the but not say,
σχεδὸν τοιαῦτα καὶ τοσαῦτά ἐστιν.
almost such and so many are.

Κεφάλαιο 7

Chapter 7

Πρὸς δὲ τὸ ποτέρως ἔχει δεῖ λαβεῖν τί σημαίνει τοὔνομα.
Regarding and the which way it is necessary to take what it means. the name.

δοκεῖ δὴ τὸ κενὸν τόπος εἶναι ἐν ᾧ μηδέν ἐστι. τούτου
It seems indeed the void to be a place to be in which nothing exists. Of this
δ' αἴτιον ὅτι τὸ ὂν σῶμα οἴονται εἶναι, πᾶν δὲ σῶμα ἐν
but the reason is that the being body they believe to be, every but body in
τόπῳ, κενὸν δὲ ἐν ᾧ τόπῳ μηδέν ἐστι σῶμα, ὥστ' εἴ που
a place, void but in which place nothing exists as a body, so that if anywhere
μὴ ἔστι σῶμα, οὐδὲν εἶναι ἐνταῦθα.
not exists body, nothing to be here.

σῶμα δὲ πάλιν ἅπαν [214a] οἴονται εἶναι ἁπτόν· τοιοῦτο δὲ

body but again all [214a] they think to be something tangible; such but

ὃ ἂν ἔχῃ βάρος ἢ κουφότητα.

what ever it may have weight or lightness.

συμβαίνει οὖν ἐκ συλλογισμοῦ τοῦτο εἶναι κενόν, ἐν ᾧ

it happens therefore out of reasoning this to be void, in which

μηδέν ἐστι βαρὺ ἢ κοῦφον.

nothing is heavy or light.

ταῦτα μὲν οὖν, ὥσπερ εἴπομεν καὶ πρότερον, ἐκ συλλογισμοῦ

these on the one hand therefore, as we said and previously, from reasoning

συμβαίνει. ἄτοπον δὲ εἰ ἡ στιγμὴ κενόν· δεῖ γὰρ τόπον

it follows. strange but if the moment is empty; it is necessary for a place

εἶναι ἐν ᾧ σώματος ἔστι διά στημα ἁπτοῦ.

to be in which of the body exists through fill of contact.

ἀλλ' οὖν φαίνεται λέγεσθαι τὸ κενὸν ἕνα μὲν τρόπον τὸ

but therefore it appears to be said the void one on the one hand in a way the

μὴ πλῆρες αἰσθητοῦ σώματος κατὰ τὴν ἁφήν· αἰσθητὸν δ'

not full of the sensible body according to the touch; sensible but

ἐστὶ κατὰ τὴν ἁφὴν τὸ βάρος ἔχον ἢ κουφότητα (διὸ κἂν

it is according to the touch the weight having or lightness (therefore even if

ἀπορήσειέ τις, τί ἂν φαῖεν, εἰ ἔχοι τὸ διάστημα χρῶμα

someone wonders, anyone, what at all they might say, if it has the expanse color

ἢ ψόφον, πότερον κενὸν ἢ οὔ; ἢ δῆλον ὅτι εἰ μὲν

or sound, which is empty or not? or it is clear that if on the one hand

δέχοιτο σῶμα ἁπτόν, κενόν, εἰ δὲ μή, οὔ)·

it were accepted body tangible, void, if but not, no)·

ἄλλον δὲ τρόπον, ἐν ᾧ μὴ τόδε τι μηδ' οὐσία τις

another but way, in which not this something and nothing substance something

σωματική. διό φασίν τινες εἶναι τὸ κενὸν τὴν τοῦ σώματος

bodily. therefore they say some to be the void the of the body

ὕλην (οἵπερ καὶ τὸν τόπον τὸ αὐτὸ τοῦτο), λέγοντες οὐ καλῶς· ἡ

matter (who also the place the same this), saying not well· the

μὲν γὰρ ὕλη οὐ χωριστὴ τῶν πραγμάτων, τὸ δὲ κενὸν ζητοῦσιν

indeed for matter not separable from the things, the but void they seek

ὡς χωριστόν.

as separate.

ἐπεὶ δὲ περὶ τόπου διώρισται, καὶ τὸ κενὸν ἀνάγκη τόπον

since but regarding place it has been defined, and the void necessity place

εἶναι, εἰ ἔστιν, ἐστερημένον σώματος, τόπος δὲ καὶ πῶς ἔστι

to be, if it exists, deprived of body, place but and how it exists

καὶ πῶς οὐκ ἔστιν εἴρηται, φανερὸν ὅτι οὕτω μὲν κενὸν οὐκ

and how not it is said, evident that thus indeed void not

ἔστιν, οὔτε κεχωρισμένον οὔτε ἀχώριστον. τὸ γὰρ κενὸν οὐ σῶμα

it is, neither separated nor inseparable. the for void not body

ἀλλὰ σώματος διάστημα βούλεται εἶναι· διὸ καὶ τὸ κενὸν

but of body space it desires to be: therefore and the void

δοκεῖ τι εἶναι, ὅτι καὶ ὁ τόπος, καὶ διὰ ταὐτά.

it seems something to be, because and the place, and through the same things.

ἥκει γὰρ δὴ ἡ κίνησις ἡ κατὰ τόπον καὶ τοῖς τὸν τόπον

it arrives for indeed the motion the according to place and those the place

φάσκουσιν εἶναί τι παρὰ τὰ σώματα τὰ ἐμπίπτοντα καὶ τοῖς

claim to be something beside the bodies the that fall and to the

τὸ κενόν. αἴτιον δὲ κινήσεως οἴονται εἶναι τὸ κενὸν οὕτως ὡς

the void. cause but of motion they think to be the void in this way as

ἐν ᾧ κινεῖται· τοῦτο δ' ἂν εἴη οἷον τὸν τόπον φασί τινες

in which is moved; this but if it were such as the place they say some

εἶναι.

to be.

οὐδεμία δ' ἀνάγκη, εἰ κίνησις ἔστιν, εἶναι κενόν. ὅλως μὲν

no but necessity, if motion exists, to be void. altogether indeed

οὖν πάσης κινήσεως οὐδαμῶς, δι' ὃ καὶ Μέλισσον

therefore of any motion, in no way, because of that which also Melissos

ἔλαθεν· ἀλλοιοῦσθαι γὰρ τὸ πλῆρες ἐνδέχεται.

escaped notice; to be altered for the full is possible.

ἀλλὰ δὴ οὐδὲ τὴν κατὰ τόπον κίνησιν· ἅμα γὰρ

but indeed neither the according to place motion; at the same time for

ἐνδέχεται ὑπεξιέναι ἀλλήλοις, οὐδενὸς ὄντος διαστήματος

it is possible to depart from one another, no being of distance

χωριστοῦ παρὰ τὰ σώματα τὰ κινούμενα. καὶ τοῦτο δῆλον καὶ ἐν

separate from the bodies the moving. And this is clear and in

ταῖς τῶν συνεχῶν δίναις, ὥσπερ καὶ ἐν ταῖς τῶν ὑγρῶν.

the of the continuous motions, just as and in the of the waters.

ἐνδέχεται δὲ καὶ πυκνοῦσθαι μὴ εἰς τὸ κενὸν ἀλλὰ διὰ τὸ

it is possible but and to become dense not into the void but through the

τὰ ἐνόντα [214b] ἐκπυρηνίζειν (οἷον ὕδατος συνθλιβομένου τὸν

the existing [214b] to combust (like water being compressed the

ἐνόντα ἀέρα), καὶ αὐξάνεσθαι οὐ μόνον εἰσιόντος τινὸς ἀλλὰ καὶ
existing air), and to increase not only entering something but also and
ἀλλοιώσει, οἷον εἰ ἐξ ὕδατος γίγνοιτο ἀήρ.
alteration, such as if from water it were to become air.

ὅλως δὲ ὅ τε περὶ τῆς αὐξήσεως λόγος καὶ τοῦ εἰς τὴν
altogether but what and concerning the increase discussion and the into the
τέφραν ἐγχεομένου ὕδατος αὐτὸς αὑτὸν ἐμποδίζει. ἢ γὰρ οὐκ
ash being poured water itself itself hinders. Or for not
αὐξάνεται ὁτιοῦν, ἢ οὐ σώματι, ἢ ἐνδέχεται δύο σώματα ἐν
it increases at all, or not by a body, or it is possible two bodies in
ταὐτῷ εἶναι (ἀπορίαν οὖν κοινὴν ἀξιοῦσι λύειν, ἀλλ' οὐ κενὸν
the same to be (a perplexity then common they assert to dissolve, but not a void
δεικνύουσιν ὡς ἔστιν), ἢ πᾶν εἶναι ἀναγκαῖον τὸ σῶμα κενόν, εἰ
they show as it is), or all to be necessary the body empty, if
πάντῃ αὐξάνεται καὶ αὐξάνεται διὰ κενοῦ. ὁ δ' αὐτὸς λόγος
in every way it increases and it increases through emptiness. the but itself word
καὶ ἐπὶ τῆς τέφρας.
and upon the ash.

ὅτι μὲν οὖν ἐξ ὧν δεικνύουσιν εἶναι τὸ κενὸν λύειν
that on one hand therefore out of what they show to be the empty to solve
ῥᾴδιον, φανερόν.
easy, evident.

Κεφάλαιο 8

Chapter 8

Ὅτι δ' οὐκ ἔστιν κενὸν οὕτω κεχωρισμένον, ὡς ἔνιοί φασι,
That but not is empty thus separated, as some say,

λέγωμεν πάλιν. εἰ γὰρ ἔστιν ἑκάστου φορά τις τῶν ἁπλῶν

let us say again. if for there is of each instance some of the simple

σωμάτων φύσει, οἷον τῷ πυρὶ μὲν ἄνω τῇ δὲ γῇ κάτω

bodies by nature, such as to the fire on one hand above the and earth below

καὶ πρὸς τὸ μέσον, δῆλον ὅτι οὐκ ἂν τὸ κενὸν αἴτιον εἴη τῆς

and towards the middle, it is clear that not would the empty cause be of the

φορᾶς. τίνος οὖν αἴτιον ἔσται τὸ κενόν; δοκεῖ γὰρ αἴτιον

motion. of whom therefore cause will be the empty thing? it seems for cause

εἶναι κινήσεως τῆς κατὰ τόπον, ταύτης δ' οὐκ ἔστιν.

to be of motion the according to place, of this but not is there.

ἔτι εἰ ἔστιν τι οἷον τόπος ἐστερημένος σώματος, ὅταν ᾖ

yet if there is something like a place deprived of a body, when it is

κενόν, ποῦ οἰσθήσεται τὸ εἰστεθὲν εἰς αὐτὸ σῶμα; οὐ γὰρ δὴ

empty, where will be perceived the placed into it body? not for indeed

εἰς ἅπαν. ὁ δ' αὐτὸς λόγος καὶ πρὸς τοὺς τὸν τόπον οἰομένους

into all. the and same argument and towards those the place thinking

εἶναί τι κεχωρισμένον, εἰς ὃν φέρεται· πῶς γὰρ οἰσθήσεται

to be something separated, into which is brought; how for will be perceived

τὸ ἐντεθὲν ἢ μενεῖ; καὶ περὶ τοῦ ἄνω καὶ κάτω καὶ περὶ

the placed or will remain? and regarding the upper and lower and regarding

τοῦ κενοῦ ὁ αὐτὸς ἁρμόσει λόγος εἰκότως· τὸ γὰρ κενὸν τόπον

the empty the same argument statement reasonably; the for empty place

ποιοῦσιν οἱ εἶναι φάσκοντες·

they create those who claim to be;

καὶ πῶς δὴ ἐνέσται ἢ ἐν [τῷ] τόπῳ ἢ ἐν τῷ κενῷ; οὐ γὰρ

and how indeed will there be or in [the] place or in the empty? not for

συμβαίνει, ὅταν ὅλον τεθῇ ὡς ἐν κεχωρισμένῳ τόπῳ καὶ

it happens, whenever whole is placed as in separated place and

ὑπομένοντι σῶμά τι· τὸ γὰρ μέρος, ἂν μὴ χωρὶς τιθῆται,

enduring body something; for the for part, if not apart is laid,

οὐκ ἔσται ἐν τόπῳ ἀλλ' ἐν τῷ ὅλῳ. ἔτι εἰ μὴ τόπος, οὐδὲ

it will not be in place but in the whole. yet if not a place, neither

κενὸν ἔσται.

void will be.

συμβαίνει δὲ τοῖς λέγουσιν εἶναι κενὸν ὡς ἀναγκαῖον, εἴπερ

it happens but to those who say to be empty as necessary, if indeed

ἔσται κίνησις, τοὐναντίον μᾶλλον, ἄν τις ἐπισκοπῇ, μὴ

there will be motion, on the contrary rather, if anyone examines, not

ἐνδέχεσθαι μηδὲ ἓν κινεῖσθαι, ἐὰν ᾖ κενόν· ὥσπερ γὰρ οἱ

to suppose nor one to move, if it is empty; just as for those

διὰ τὸ ὅμοιον φάμενοι τὴν γῆν ἠρεμεῖν, οὕτω καὶ ἐν τῷ

through the similar saying the earth to be at rest, so and in the

κενῷ ἀνάγκη ἠρεμεῖν· οὐ γὰρ ἔστιν οὗ μᾶλλον ἢ ἧττον

void necessity to be at rest; not for there is of which rather than less

κινηθήσεται· ᾗ γὰρ κενόν, οὐκ [215a] ἔχει διαφοράν.

it will move in which for empty, not [215a] has difference.

ἔπειθ' ὅτι πᾶσα κίνησις ἢ βίᾳ ἢ κατὰ φύσιν. ἀνάγκη δὲ

It is evident that every motion or by force or according to nature. Necessity but

ἄν περ ᾖ <ἡ> βίαιος, εἶναι καὶ τὴν κατὰ φύσιν (ἡ μὲν

if indeed it is the violent, to be and the according to nature (the one

γὰρ βίαιος παρὰ φύσιν, ἡ δὲ παρὰ φύσιν ὑστέρα

for violent (force) contrary to nature, the while contrary to nature (is later

τῆς κατὰ φύσιν)· ὥστ' εἰ μὴ κατὰ φύσιν ἔστιν ἑκάστῳ

than according to nature). Thus, if not according to nature there is for each

τῶν φυσικῶν σωμάτων κίνησις, οὐδὲ τῶν ἄλλων ἔσται κινήσεων

of the natural bodies motion, nor of the others will be motions

οὐδεμία. ἀλλὰ μὴν φύσει γε πῶς ἔσται μηδεμιᾶς οὔσης

none. But indeed by nature certainly how will be of no existing

διαφορᾶς κατὰ τὸ κενὸν καὶ τὸ ἄπειρον; ᾗ μὲν γὰρ

difference according to the void and the infinite? In what for indeed

ἄπειρον, οὐδὲν ἔσται ἄνω οὐδὲ κάτω οὐδὲ μέσον, ᾗ δὲ

the infinite, nothing will be above nor below nor middle, in which while

κενόν, οὐδὲν διάφορον τὸ ἄνω τοῦ κάτω (ὥσπερ γὰρ τοῦ μηδενὸς

void, nothing different the upper of the lower (as for of the nothing

οὐδεμία ἔστι διαφορά, οὕτω καὶ τοῦ κενοῦ· τὸ γὰρ κενὸν μὴ ὄν

no is difference, thus and of the void; the for void not being

τι καὶ στέρησις δοκεῖ εἶναι). ἡ δὲ φύσει φορὰ διάφορος,

anything and deprivation seems to be). the but by nature movement different,

ὥστε ἔσται φύσει διάφορα. ἢ οὖν οὐκ ἔστι φύσει οὐδαμοῦ

so that it will be by nature different. Or then not is by nature nowhere

οὐδενὶ φορά, ἢ εἰ τοῦτ' ἔστιν, οὐκ ἔστι κενόν.

to nothing movement, or if this is, not is void.

ἔτι νῦν μὲν κινεῖται τὰ ῥιπτούμενα τοῦ ὤσαντος οὐχ ἁπτομένου,

yet now indeed is moved the thrown of the moving not being touched,

ἢ δι' ἀντιπερίστασιν, ὥσπερ ἔνιοί φασιν, ἢ διὰ τὸ ὠθεῖν τὸν

or by opposition, as some say, or through the pushing the

ὠσθέντα ἀέρα θάττω κίνησιν τῆς τοῦ ὠσθέντος φορᾶς ἣν

pushed air faster motion of the the pushed movement which

φέρεται εἰς τὸν οἰκεῖον τόπον· ἐν δὲ τῷ κενῷ οὐδὲν τούτων

is brought into the proper place. In but the void nothing of these

ὑπάρχει, οὐδ' ἔσται φέρεσθαι ἀλλ' ἢ ὡς τὸ ὀχούμενον.

exists, nor will be brought but rather as the being carried.

ἔτι οὐδεὶς ἂν ἔχοι εἰπεῖν διὰ τί κινηθὲν στήσεταί

still no one if should have to say because of why having been moved will be set

που· τί γὰρ μᾶλλον ἐνταῦθα ἢ ἐνταῦθα; ὥστε ἢ ἠρεμήσει

somewhere; what for rather here than here? so that or will be at rest

ἢ εἰς ἄπειρον ἀνάγκη φέρεσθαι, ἐὰν μή τι ἐμποδίσῃ κρεῖττον.

or into infinity necessity to be carried, if not anything should hinder better.

ἔτι νῦν μὲν εἰς τὸ κενὸν διὰ τὸ ὑπείκειν φέ ρεσθαι

still now on one hand into the void through the yielding to bear to be carried

δοκεῖ· ἐν δὲ τῷ κενῷ πάντῃ ὁμοίως τὸ τοιοῦτον, ὥστε πάντῃ

it seems; in but the void everywhere similarly the such, so that everywhere

οἰσθήσεται. ἔτι δὲ καὶ ἐκ τῶνδε φανερὸν τὸ λεγόμενον. ὁρῶμεν

it will be sensed. still but also from these clear the said. we see

γὰρ τὸ αὐτὸ βάρος καὶ σῶμα θᾶττον φερόμενον διὰ δύο αἰτίας,

for the same weight and body sooner being carried through two causes,

ἢ τῷ διαφέρειν τὸ δι' οὗ, οἷον δι' ὕδατος ἢ γῆς ἢ δι'

or by the differing the through which, such as through water or of earth or by

ὕδατος ἢ ἀέρος, ἢ τῷ διαφέρειν τὸ φερόμενον, ἐὰν τἆλλα ταὐτὰ

water or air, or to the differences the being carried, if the others the same

ὑπάρχῃ, διὰ τὴν ὑπεροχὴν τοῦ βάρους ἢ τῆς κουφότητος.

exists, through the superiority of the weight or of the lightness.

τὸ μὲν οὖν δι' οὗ φέρεται αἴτιον, ὅτι ἐμποδίζει

the on the one hand therefore by which is brought cause, because hinders

μάλιστα μὲν ἀντιφερόμενον, ἔπειτα καὶ μένον· μᾶλλον

particularly on the one hand opposing, afterwards and remaining; rather

δὲ τὸ μὴ εὐδιαίρετον· τοιοῦτο δὲ τὸ παχύτερον. τὸ δὴ ἐφ' οὗ

but the not easily divisible; such but the thicker. the indeed upon which

[215b] A οἰσθήσεται διὰ τοῦ B τὸν ἐφ' ᾧ Γ χρόνον, διὰ

[215b] A will be perceived through the B the upon which C time, through

δὲ τοῦ Δ λεπτοτέρου ὄντος τὸν ἐφ' ᾧ E, εἰ ἴσον τὸ μῆκος

but the D thinner being the upon which E, if equal the length

τὸ τοῦ B τῷ Δ, κατὰ τὴν ἀναλογίαν τοῦ ἐμποδίζοντος

the of the B to the D, according to the proportion of the hindering

σώματος. ἔστω γὰρ τὸ μὲν B ὕδωρ, τὸ δὲ Δ ἀήρ·

body. let it be for the on the one hand B water, the but D air;

ὅσῳ δὴ λεπτότερον ἀὴρ ὕδατος καὶ ἀσωματώτερον, τοσούτῳ

to the extent indeed thinner air of water and less material. to such an extent

θᾶττον τὸ A διὰ τοῦ Δ οἰσθήσεται ἢ διὰ τοῦ B. ἐχέτω

quicker the A through of the D will be known or through of the B. let there be

δὴ τὸν αὐτὸν λόγον ὅνπερ διέστηκεν ἀὴρ πρὸς ὕδωρ,

indeed the same account which has been distinguished air towards water,

τὸ τάχος πρὸς τὸ τάχος. ὥστε εἰ διπλασίως λεπτόν, ἐν διπλασίῳ

the speed towards the speed. so that if twice thin, in double

χρόνῳ τὴν τὸ B δίεισιν ἢ τὴν τὸ Δ, καὶ ἔσται ὁ ἐφ' ᾧ Γ

time the the B passes through or the the D, and will be the upon which G

χρόνος διπλάσιος τοῦ ἐφ' ᾧ E. καὶ ἀεὶ δὴ ὅσῳ ἂν ᾖ

time double of the upon which E. and always indeed as much as ever is

ἀσωματώτερον καὶ ἧττον ἐμποδιστικὸν καὶ εὐδιαιρετώτερον δι'

more incorporeal and less obstructive and more divisible through

οὗ φέρεται, θᾶττον οἰσθήσεται.

which is carried, quicker will be known.

τὸ δὲ κενὸν οὐδένα ἔχει λόγον ᾧ ὑπερέχεται ὑπὸ τοῦ σώματος,

the but empty no one has reason for which it is surpassed by the body,

ὥσπερ οὐδὲ τὸ μηδὲν πρὸς ἀριθμόν. εἰ γὰρ τὰ τέτταρα τῶν τριῶν

just as nor the nothing towards number. if for the four of three

ὑπερέχει ἑνί, πλείονι δὲ τοῖν δυοῖν, καὶ ἔτι πλείονι τοῦ ἑνὸς ἢ

surpasses one, more but of the two, and yet more of the one or

τοῖν δυοῖν, τοῦ δὲ μηδενὸς οὐκέτι ἔχει λόγον ᾧ ὑπερέχει·

of the two, of the but of nothing no longer has reason for which it surpasses;

ἀνάγκη γὰρ τὸ ὑπερέχον διαιρεῖσθαι εἴς τε τὴν ὑπεροχὴν καὶ τὸ

necessity for the surpassing to be divided into both the superiority and the

ὑπερεχόμενον, ὥστε ἔσται τὰ τέτταρα ὅσῳ τε ὑπερέχει καὶ

surpassed, so that it will be the four as much as and it surpasses and

οὐδέν. διὸ οὐδὲ γραμμὴ στιγμῆς ὑπερέχει, εἰ μὴ σύγκειται ἐκ

nothing. therefore nor a line of a point surpasses, if not composed of

στιγμῶν. ὁμοίως δὲ καὶ τὸ κενὸν πρὸς τὸ πλῆρες οὐδένα οἷόν

points. similarly but also the empty towards the full no one such

τε ἔχειν λόγον, ὥστε οὐδὲ τὴν κίνησιν, ἀλλ' εἰ διὰ τοῦ

and to have reason, so that not even the motion, but if through the

λεπτοτάτου ἐν τοσῳδὶ τὴν τοσήνδε φέρεται, διὰ τοῦ κενοῦ

most subtle in so great the such is carried, through the void

παντὸς ὑπερβάλλει λόγου. ἔστω γὰρ τὸ Z κενόν, ἴσον δὲ [τῷ

of all exceeds reason. let it be for the Z void, equal but to the

μεγέθει] τοῖς B καὶ Δ. τὸ δὴ A εἰ δίεισι καὶ κινηθήσεται

size] to the B and D. the indeed A if it passes and will be set in motion

ἐν τινὶ μὲν χρόνῳ, τῷ ἐφ' οὗ H, ἐν ἐλάττονι δὲ τοῦ ἐφ'

in some certain time, the at which it H, in lesser but of the at which

οὗ Ε, τοῦτον ἕξει τὸν λόγον τὸ κενὸν πρὸς τὸ πλῆρες. ἀλλ'

it E, this will have the reason the void towards the full. but

ἐν τοσούτῳ χρόνῳ ὅσος ἐφ' οὗ τὸ Η, τοῦ Δ τὸ Α δίεισι

in such a time as much as at which it the The, of the D the A passes

τὴν τὸ Θ. δίεισι δέ γε κἂν ᾖ τι λεπτότητι διαφέρον τοῦ

the the θ. passes but indeed even if it is something thinness differing of the

ἀέρος ἐφ' ᾧ τὸ Ζ ταύτην τὴν ἀναλογίαν ἣν ἔχει ὁ χρόνος ἐφ'

air upon which the Z this the analogy which has the time upon

ᾧ Ε πρὸς τὸν ἐφ' ᾧ Η. ἂν γὰρ ᾖ τοσούτῳ λεπτότερον

which E towards the upon which H. if for it is so much thinner

τὸ ἐφ' ᾧ Ζ σῶμα τοῦ Δ, ὅσῳ ὑπερέχει τὸ Ε τοῦ Η,

the upon which Z body of the D, as much as exceeds the E of the H,

ἀντεστραμμένως δίεισι τῷ [216a] τάχει ἐν τῷ τοσούτῳ ὅσον

in reverse passes to the [216a] speed in the so much as

τὸ Η, τὴν τὸ Ζ τὸ ἐφ' οὗ Α, ἐὰν φέρηται. ἐὰν τοίνυν μηδὲν

the H, the the Z the upon of which A, if it is brought. if therefore nothing

ᾖ σῶμα ἐν τῷ Ζ, ἔτι θᾶττον. ἀλλ' ἦν ἐν τῷ Η. ὥστ' ἐν ἴσῳ

it is body in the Z, yet sooner. but it was in the H. so that in equal

χρόνῳ δίεισι πλῆρές τε ὂν καὶ κενόν. ἀλλ' ἀδύνατον. φανερὸν

time it passes full and being and empty. but impossible. evident

τοίνυν ὅτι, εἰ ἔστι χρόνος ἐν ᾧ τοῦ κενοῦ ὁτιοῦν οἰσθήσεται,

therefore that, if it is time in which of the empty anything will be perceived,

συμβήσεται τοῦτο τὸ ἀδύνατον· ἐν ἴσῳ γὰρ ληφθήσεται πλῆρές τε

it will happen this the impossible; in equal for it will be taken full and

ὂν διεξιέναι τι καὶ κενόν· ἔσται γάρ τι ἀνάλογον

being to pass through something and empty; it will be for something analogous

σῶμα ἕτερον πρὸς ἕτερον ὡς χρόνος πρὸς χρόνον.

body another towards another as time towards time.

ὡς δ’ ἐν κεφαλαίῳ εἰπεῖν, δῆλον τὸ τοῦ συμβαίνοντος αἴτιον, ὅτι

as but in head to say, clear the of the happening cause, that

κινήσεως μὲν πρὸς κίνησιν πάσης ἔστι λόγος (ἐν χρόνῳ

of movement on the one hand towards motion of all is a reason (in time

γάρ ἐστι, χρόνου δὲ παντὸς ἔστι πρὸς χρόνον, πεπερασμένων

for is, of time but of all is towards time, of the finite

ἀμφοῖν), κενοῦ δὲ πρὸς πλῆρες οὐκ ἔστιν. ᾗ μὲν

both), of the void but towards the full not is. In which on the one hand

οὖν διαφέρουσι δι’ ὧν φέρονται, ταῦτα συμβαίνει, κατὰ

therefore they differ because of which are carried, these happen, according to

δὲ τὴν τῶν φερομένων ὑπεροχὴν τάδε· ὁρῶμεν γὰρ τὰ μείζω

but the of the carried excess these: we see for the greater

ῥοπὴν ἔχοντα ἢ βάρους ἢ κουφότητος, ἐὰν τἆλλα ὁμοίως ἔχῃ

weight having or of weight or of lightness, if the others likewise has

[τοῖς σχήμασι], θᾶττον φερόμενα τὸ ἴσον χωρίον, καὶ κατὰ

[the forms], faster being carried the equal place, and according to

λόγον ὃν ἔχουσι τὰ μεγέθη πρὸς ἄλληλα. ὥστε καὶ διὰ

the reason which they have the magnitudes towards mutually. so that and through

τοῦ κενοῦ. ἀλλ’ ἀδύνατον· διὰ τίνα γὰρ αἰτίαν οἰσθήσεται

the void. but impossible; through what indeed reason it will be perceived

θᾶττον; ἐν μὲν γὰρ τοῖς πλήρεσιν ἐξ ἀνάγκης· θᾶττον

sooner? in on the one hand indeed the fullest than necessity; sooner

γὰρ διαιρεῖ τῇ ἰσχύϊ τὸ μεῖζον· ἢ γὰρ σχήματι διαιρεῖ,

indeed it divides by the force the greater; or indeed by the shape it divides,

ἢ ῥοπῇ ἣν ἔχει τὸ φερόμενον ἢ τὸ ἀφεθέν. ἰσοταχῆ

or by the tendency which it has the carried or the released. equal-speed

ἄρα πάντ' ἔσται. ἀλλ' ἀδύνατον. ὅτι μὲν οὖν εἰ ἔστι κενόν,

thus all will be. but impossible. that indeed therefore if it is empty,

συμβαίνει τοὐναντίον ἢ δι' ὃ κατασκευάζουσιν οἱ φάσκοντες

it follows the opposite or through which they construct the claimers

εἶναι κενόν, φανερὸν ἐκ τῶν εἰρημένων. οἱ μὲν οὖν οἴονται

to be empty, evident from the aforementioned. they indeed therefore believe

τὸ κενὸν εἶναι, εἴπερ ἔσται ἡ κατὰ τόπον κίνησις, ἀποκεκριμένον

the empty to be, if ever it will be the according to place motion, having been separated

καθ' αὑτό· τοῦτο δὲ ταὐτόν ἐστι τῷ τὸν τόπον φάναι εἶναί

according to itself; this but the same is to the the place to say to be

τι κεχωρισμένον· τοῦτο δ' ὅτι ἀδύνατον, εἴρηται

something separated; this but that it is impossible, it has been said

πρότερον. καὶ καθ' αὑτὸ δὲ σκοποῦσιν φανείη

before. And according to itself but they are contemplating it would appear

ἂν τὸ λεγόμενον κενὸν ὡς ἀληθῶς κενόν. ὥσπερ γὰρ ἐὰν ἐν

if the said is empty as truly empty. Just as for if in

ὕδατι τιθῇ τις κύβον, ἐκστήσεται τοσοῦτον ὕδωρ ὅσος

water someone places a cube, it will displace so much water as much as

ὁ κύβος, οὕτω καὶ ἐν ἀέρι· ἀλλὰ τῇ αἰσθήσει ἄδηλον.

the cube, so and in air; but to the perception it is unclear.

καὶ αἰεὶ δὴ ἐν παντὶ σώματι ἔχοντι μετάστασιν, ἐφ' ὃ

And always indeed in every body having a change, upon which that

πέφυκε μεθίστασθαι, ἀνάγκη, ἂν μὴ συμπιλῆται,

it has naturally come to change, it is necessary, if not it is pressed together,

μεθίστασθαι ἢ κάτω αἰεί, εἰ κάτω ἡ φορὰ ὥσπερ γῆς, ἢ ἄνω,

to change or downwards always, if down the motion just as of earth, or above,

εἰ πῦρ, ἢ ἐπ' ἄμφω, [ἢ] ὁποῖον ἄν τι ᾖ τὸ ἐντιθέμενον·

if fire, or upon both, [or] whatever may anything may be the that which is placed;

ἐν δὲ δὴ τῷ κενῷ τοῦτο μὲν ἀδύνατον (οὐδὲν γὰρ
in but indeed in the void this on the one hand is impossible (nothing for
σῶμα), διὰ δὲ τοῦ κύβου τὸ ἴσον διάστημα διεληλυθέναι,
body), because of but the cube the equal distance has been traversed,
ὅπερ ἦν καὶ πρότερον [216b] ἐν τῷ κενῷ, ὥσπερ ἂν εἰ τὸ ὕδωρ
which was and before [216b] in the void, just as may if the water
μὴ μεθίστατο τῷ ξυλίνῳ κύβῳ μηδ' ὁ ἀήρ, ἀλλὰ πάντῃ
not change with the wooden cube nor the air, but everywhere
διήεσαν δι' αὐτοῦ. ἀλλὰ μὴν καὶ ὁ κύβος γε ἔχει
they would be separated by it. But indeed also the cube surely has
τοσοῦτον μέγεθος, ὅσον κατέχει κενόν· ὃ εἰ καὶ θερμὸν ἢ
so much size, as it occupies a void; which if even hot or
ψυχρόν ἐστιν ἢ βαρὺ ἢ κοῦφον, οὐδὲν ἧττον ἕτερον τῷ εἶναι
cold it is or heavy or light, nothing less different to the being
πάντων τῶν παθημάτων ἐστί, καὶ εἰ μὴ χωριστόν· λέγω δὲ τὸν
of all the affections is, and if not separate; I say but the
ὄγκον τοῦ ξυλίνου κύβου. ὥστ' εἰ καὶ χωρισθείη τῶν ἄλλων
volume of the wooden cube. so that if also it were separated from the others
πάντων καὶ μήτε βαρὺ μήτε κοῦφον εἴη, καθέξει τὸ ἴσον
all and neither heavy nor light it would be, it would contain the equal
κενὸν καὶ ἐν τῷ αὐτῷ ἔσται τῷ τοῦ τόπου καὶ τῷ τοῦ
void and in the same it will be to the of the place and to the of the
κενοῦ μέρει ἴσῳ ἑαυτῷ. τί οὖν διοίσει τὸ τοῦ κύβου σῶμα
void part equal to itself. What then will divide the of the cube body
τοῦ ἴσου κενοῦ καὶ τόπου; καὶ εἰ δύο τοιαῦτα, διὰ τί οὐ καὶ
of the equal void and place? And if two such, through what not also
ὁποσαοῦν ἐν τῷ αὐτῷ ἔσται;
any in the same will exist?

ἓν μὲν δὴ τοῦτο ἄτοπον καὶ ἀδύνατον. ἔτι δὲ φανερὸν
one indeed truly this absurd and impossible. furthermore but evident
ὅτι τοῦτο ὁ κύβος ἕξει καὶ μεθιστάμενος, ὃ καὶ τὰ ἄλλα
that this the cube will have and being relocated, which also the other
σώματα πάντ' ἔχει. ὥστ' εἰ τοῦ τόπου μηδὲν διαφέρει, τί δεῖ
bodies all have. therefore if of the place nothing differs, what need
ποιεῖν τόπον τοῖς σώμασιν παρὰ τὸν ἑκάστου ὄγκον, εἰ ἀπαθὲς
to do place to the bodies besides the of each bulk, if unchangeable
ὁ ὄγκος; οὐδὲν γὰρ συμ βάλλεται, εἰ ἕτερον περὶ αὐτὸν ἴσον
the bulk; nothing for it is thrown, if another regarding it equal
διάστημα τοιοῦτον εἴη. [ἔτι δεῖ δῆλον εἶναι οἷον κενὸν
distance such might be. [furthermore need clear to be such as void
ἐν τοῖς κινουμένοις. νῦν δ' οὐδαμοῦ ἐντὸς τοῦ κόσμου· ὁ γὰρ
in the moving. now but nowhere within the universe; the for
ἀὴρ ἔστιν τι, οὐ δοκεῖ δέ γε-οὐδὲ τὸ ὕδωρ, εἰ ἦσαν οἱ
air is something, not it seems but indeed not even the water, if they were the
ἰχθύες σιδηροῖ· τῇ ἁφῇ γὰρ ἡ κρίσις τοῦ ἁπτοῦ.]
fish of iron. by the touch for the judgment of the tangible.]

ὅτι μὲν τοίνυν οὐκ ἔστι κεχωρισμένον κενόν, ἐκ τούτων
that on the one hand therefore not is separated empty, from these
ἐστὶ δῆλον.
it is clear.

Κεφάλαιο 9

Chapter 9

Εἰσὶν δέ τινες οἳ διὰ τοῦ μανοῦ καὶ πυκνοῦ οἴονται φανερὸν

They are but some who through the of the mind and dense think evident

εἶναι ὅτι ἔστι κενόν. εἰ μὲν γὰρ μὴ ἔστι μανὸν καὶ πυκνόν,

to be that it is empty. if indeed for not it is of the mind and dense,

οὐδὲ συνιέναι καὶ πιλεῖσθαι οἷόν τε· εἰ δὲ τοῦτο μὴ εἴη,

neither to understand and to be filled such as (also); if but this not should be,

ἢ ὅλως κίνησις οὐκ ἔσται, ἢ κυμανεῖ τὸ ὅλον, ὥσπερ

or entirely movement not will there be, or it will change the whole, just as

ἔφη Ξοῦθος, ἢ εἰς ἴσον ἀεὶ <δεῖ> μεταβάλλειν ἀέρα καὶ ὕδωρ

said Xouthos, or into the same always <must> to change air and water

(λέγω δὲ οἷον εἰ ἐξ ὕδατος κυάθου γέγονεν ἀήρ, ἅμα ἐξ

(I say but as if if out of water of a cup has become air, together out of

ἴσου ἀέρος ὕδωρ τοσοῦτον γεγενῆσθαι), ἢ κενὸν εἶναι ἐξ

the same air water so much to have become), or empty to be out of

ἀνάγκης· συμπιλεῖσθαι γὰρ καὶ ἐπεκτείνεσθαι οὐκ ἐνδέχεται

necessity; to be filled for and to extend not it is possible

ἄλλως.

otherwise.

εἰ μὲν οὖν τὸ μανὸν λέγουσι τὸ πολλὰ κενὰ κεχωρισμένα

if indeed therefore the mind they say the many empty separated

ἔχον, φανερὸν ὡς εἰ μηδὲ κενὸν ἐνδέχεται εἶναι χωριστὸν ὥσπερ

having, it is clear that if neither empty it may be to be separate just as

μηδὲ τόπον ἔχοντα διάστημα αὑτοῦ, οὐδὲ μανὸν οὕτως· εἰ δὲ

neither place having distance of itself, nor mind thus; if but

μὴ χωριστόν, ἀλλ' ὅμως ἐνεῖναί τι κενόν, ἧττον μὲν

not separated, but nonetheless it must be something empty, less indeed

ἀδύνατον, συμβαίνει δὲ πρῶτον μὲν οὐ πάσης κινήσεως αἴτιον τὸ

impossible, it happens but first indeed not of every motion cause the

κενόν, ἀλλὰ τῆς ἄνω (τὸ γὰρ μανὸν κοῦφον, διὸ [217a] καὶ

empty, but of the upper (the for mind is light, therefore [217a] and

τὸ πῦρ μανὸν εἶναί φασιν), ἔπειτα κινήσεως αἴτιον οὐχ οὕτω τὸ

the fire is mind to be they say), then of motion cause not so the

κενὸν ὡς ἐν ᾧ, ἀλλ' ὥσπερ οἱ ἀσκοὶ τῷ φέρεσθαι αὐτοὶ ἄνω

empty as in which, but just as the bags to the carry they upwards

φέρουσι τὸ συνεχές, οὕτω τὸ κενὸν ἄνω φέρει. καίτοι πῶς οἷόν

carry the continuous, thus the empty upwards carries. And yet how possible

τε φορὰν εἶναι κενοῦ ἢ τόπον κενοῦ; κενοῦ γὰρ γίγνεται

and a bearing to be of emptiness or a place of emptiness? of emptiness for it becomes

κενόν, εἰς ὃ φέρεται.

empty, into which it is carried.

ἔτι δὲ πῶς ἐπὶ τοῦ βαρέος ἀποδώσουσιν τὸ φέρεσθαι κάτω; καὶ

yet but how upon the heavy will they return the to be carried down? and

δῆλον ὅτι εἰ ὅσῳ ἂν μανότερον καὶ κενώτερον ᾖ ἄνω

clear that if to the extent at all more full and more empty may be above

οἰσθήσεται, εἰ ὅλως εἴη κενόν, τάχιστ' ἂν φέροιτο.

it will be felt, if altogether it were empty, as quickly as it might be carried.

ἴσως δὲ καὶ τοῦτ' ἀδύνατον κινηθῆναι· λόγος δ' ὁ αὐτός,

perhaps but also this cannot be moved; reasoning but the same,

ὥσπερ ὅτι ἐν τῷ κενῷ ἀκίνητα πάντα, οὕτω καὶ τὸ κενὸν ὅτι

just as that in the void immobile all, so and the void that

ἀκίνητον· ἀσύμβλητα γὰρ τὰ τάχη. ἐπεὶ δὲ κενὸν μὲν οὔ

it is immobile; incompatible for the speeds. since but void on the one hand not

φαμεν εἶναι, τὰ ἄλλα δ' ἠπόρηται ἀληθῶς, ὅτι ἢ κίνησις οὐκ

we say to be, the others but are puzzled truly, that either motion will not

ἔσται, εἰ μὴ ἔσται πύκνωσις καὶ μάνωσις, ἢ κυμανεῖ ὁ

exist, if not there will be a densification and a thinning, or will it move the

οὐρανός, ἢ αἰεὶ ἴσον ὕδωρ ἐξ ἀέρος ἔσται καὶ ἀὴρ ἐξ ὕδατος

sky, or always equal water from air will be and air from water

(δῆλον γὰρ ὅτι πλείων ἀὴρ ἐξ ὕδατος γίγνεται· ἀνάγκη τοίνυν,

(evidently for that more air from water arises· necessity therefore,

εἰ μὴ ἔστι πίλησις, ἢ ἐξωθούμενον τὸ ἐχόμενον τὸ ἔσχατον

if not there is a compression, or pushed out the contained the last

κυμαίνειν ποιεῖν, ἢ ἄλλοθί που ἴσον μεταβάλλειν ἐξ ἀέρος

to wave, to make, or from elsewhere somewhere equal to change from air

ὕδωρ, ἵνα ὁ πᾶς ὄγκος τοῦ ὅλου ἴσος ᾖ, ἢ μηδὲν κινεῖσθαι·

water, so that the whole mass of the whole equal may be, or nothing to be moved·

ἀεὶ γὰρ μεθισταμένου τοῦτο συμβήσεται, ἂν μὴ κύκλῳ περιίστηται·

always for of the changing, this will happen, if not in a circle it is contained·

οὐκ ἀεὶ δ' εἰς τὸ κύκλῳ ἡ φορά, ἀλλὰ καὶ εἰς εὐθύ)·

not always but into the circle the motion, but also into straight)·

οἱ μὲν δὴ διὰ ταῦτα κενόν τι φαῖεν ἂν εἶναι, ἡμεῖς δὲ

the indeed indeed through these empty something might say if to be, we but

λέγομεν ἐκ τῶν ὑποκειμένων ὅτι ἔστιν ὕλη μία τῶν ἐναντίων,

say from the subjects that there is matter one of the opposites,

θερμοῦ καὶ ψυχροῦ καὶ τῶν ἄλλων τῶν φυσικῶν ἐναντιώσεων, καὶ

of hot and cold and of the other of the natural oppositions, and

ἐκ δυνάμει ὄντος ἐνεργείᾳ ὂν γίγνεται, καὶ οὐ χωριστὴ μὲν ἡ

from power being action being becomes, and not separate indeed the

ὕλη, τὸ δ' εἶναι ἕτερον, καὶ μία τῷ ἀριθμῷ, εἰ ἔτυχε, χροιᾶς
matter, the but to be another, and one in the number, if it happened, of colors
καὶ θερμοῦ καὶ ψυχροῦ.
and of hot and cold.

ἔστι δὲ καὶ σώματος ὕλη καὶ μεγάλου καὶ μικροῦ ἡ αὐτή.
there is but also of body matter and of large and small the same.
δῆλον δέ· ὅταν γὰρ ἐξ ὕδατος ἀὴρ γένηται, ἡ αὐτὴ ὕλη οὐ
clear indeed; when for from water air becomes, the same matter not
προσλαβοῦσά τι ἄλλο ἐγένετο, ἀλλ' ὃ ἦν δυνάμει, ἐνεργείᾳ
having taken anything else became, but that it was in power, in activity
ἐγένετο, καὶ πάλιν ὕδωρ ἐξ ἀέρος ὡσαύτως, ὁτὲ μὲν εἰς
became, and again water out of air likewise, sometimes indeed into
μέγεθος ἐκ μικρότητος, ὁτὲ δ' εἰς μικρό τητα ἐκ μεγέθους.
greatness from smallness, sometimes but into small size from greatness.
ὁμοίως τοίνυν κἂν ὁ ἀὴρ πολὺς ὢν ἐν ἐλάττονι γίγνηται ὄγκῳ
similarly therefore even if the air much being in lesser becomes in bulk
καὶ ἐξ ἐλάττονος μείζων, ἡ δυνάμει οὖσα ὕλη γίγνεται ἄμφω.
and out of lesser greater, the power being matter becomes both.

ὥσπερ γὰρ καὶ ἐκ ψυχροῦ θερμὸν καὶ ἐκ θερμοῦ ψυχρὸν ἡ
just as for and out of cold hot and out of warm cold the
αὐτή, ὅτι ἦν δυνάμει, οὕτω καὶ ἐκ θερμοῦ [217b] μᾶλλον
same, because it was in power, so and out of warm [217b] more
θερμόν, οὐδενὸς γενομένου ἐν τῇ ὕλῃ θερμοῦ ὃ οὐκ ἦν θερμὸν
warm, of none becoming in the wood warm which not was hot
ὅτε ἧττον ἦν θερμόν, ὥσπερ γε οὐδ' ἡ τοῦ μείζονος κύκλου
when lesser was hot, just as indeed nor the of the greater circle

περιφέρεια καὶ κυρτότης ἐὰν γίγνηται ἐλάττονος κύκλου, <ἢ>

circumference and curvature if it becomes lesser of the circle, <or>

ἡ αὐτὴ οὖσα ἢ ἄλλη, ἐν οὐθενὶ ἐγγέγονε τὸ κυρτὸν ὃ

the same being or another, in none has been produced the curved which

ἦν οὐ κυρτὸν ἀλλ' εὐθύ (οὐ γὰρ τῷ διαλείπειν τὸ ἧττον ἢ τὸ

was not curved but straight (not for the to interrupt the lesser or the

μᾶλλον ἔστιν)· οὐδ' ἔστι τῆς φλογὸς λαβεῖν τι μέγεθος ἐν

more is)· nor is of the flame to take some size in

ᾧ οὐ καὶ θερμότης καὶ λευκότης ἔνεστιν. οὕτω τοίνυν καὶ ἡ

which not and warmth and whiteness is present. thus therefore and the

πρότερον θερμότης <πρὸς> τὴν ὕστερον. ὥστε καὶ τὸ μέγεθος καὶ

former warmth towards the latter. so that and the size and

ἡ μικρότης τοῦ αἰσθητοῦ ὄγκου οὐ προσλαβούσης τι τῆς

the smallness of the perceptible volume not having added anything of the

ὕλης ἐπεκτείνεται, ἀλλ' ὅτι δυνάμει ἐστὶν ὕλη ἀμφοῖν· ὥστ'

substance extends, but because potentially is matter for both. so

ἐστὶ τὸ αὐτὸ πυκνὸν καὶ μανόν, καὶ μία ὕλη αὐτῶν. ἔστι δὲ

it is the same dense and rare, and one substance of them. it is but

τὸ μὲν πυκνὸν βαρύ, τὸ δὲ μανὸν κοῦφον. [ἔτι ὥσπερ ἡ τοῦ

the one dense heavy, the other rare light. [still just as the of the

κύκλου περιφέρεια συναγομένη εἰς ἔλαττον οὐκ ἄλλο τι λαμβάνει

circle circumference gathered into lesser not another thing takes

τὸ κοῖλον, ἀλλ' ὃ ἦν συνήχθη, καὶ τοῦ πυρὸς ὅ τι ἄν

the hollow, but which was gathered and of the fire what anything at all

τις λάβῃ πᾶν ἔσται θερμόν, οὕτω καὶ τὸ πᾶν συναγωγὴ καὶ

someone should take all will be warm, thus and the whole gathering and

διαστολὴ τῆς αὐτῆς ὕλης.] δύο γὰρ ἔστιν ἐφ' ἑκατέρου, τοῦ τε

distinction of the same matter.] Two for there is on each side, of the and

πυκνοῦ καὶ τοῦ μανοῦ· τό τε γὰρ βαρὺ καὶ τὸ σκληρὸν πυκνὰ

dense and of the rare; the and for heavy and the hard densely

δοκεῖ εἶναι, καὶ τἀναντία μανὰ τό τε κοῦφον καὶ τὸ μαλακόν·

seems to be, and the opposite rare the and light and the soft;

διαφωνεῖ δὲ τὸ βαρὺ καὶ τὸ σκληρὸν ἐπὶ μολίβδου καὶ σιδήρου.

differs but the heavy and the hard upon lead and iron.

ἐκ δὴ τῶν εἰρημένων φανερὸν ὡς οὔτ' ἀποκεκριμένον κενὸν

from indeed the aforementioned evident that neither set apart void

ἔστιν, οὔθ' ἁπλῶς οὔτ' ἐν τῷ μανῷ, οὔτε δυνάμει, εἰ μή τις

is, nor simply nor in the matter, nor by power, if not anyone

βούλεται πάντως καλεῖν κενὸν τὸ αἴτιον τοῦ φέρεσθαι. οὕτω

wishes at all to call void the cause of the being carried. Thus

δ' ἡ τοῦ βαρέος καὶ κούφου ὕλη, ᾗ τοιαύτη, εἴη ἂν

but the of the heavy and light substance, which is such, would be if

τὸ κενόν· τὸ γὰρ πυκνὸν καὶ τὸ μανὸν κατὰ ταύτην τὴν

the void; the for dense and the thin according to this the

ἐναντίωσιν φορᾶς ποιητικά, κατὰ δὲ τὸ σκληρὸν καὶ μαλακὸν

opposition of forces creatively, according to but the hard and soft

πάθους καὶ ἀπαθείας, καὶ οὐ φορᾶς ἀλλ' ἑτεροιώσεως μᾶλλον.

of emotions and of apathy, and not of forces but of difference more.

καὶ περὶ μὲν κενοῦ, πῶς ἔστι καὶ πῶς οὐκ ἔστι, διωρίσθω

And regarding indeed void, how it is and how not is, let it be defined

τὸν τρόπον τοῦτον.

the manner this.

Κεφάλαιο 10

Chapter 10

Ἐχόμενον δὲ τῶν εἰρημένων ἐστὶν ἐπελθεῖν περὶ χρόνου·

Related and of the aforementioned is to approach regarding time.

πρῶτον δὲ καλῶς ἔχει διαπορῆσαι περὶ αὐτοῦ καὶ διὰ τῶν

First and well it is to inquire about it and through the

ἐξωτερικῶν λόγων, πότερον τῶν ὄντων ἐστὶν ἢ τῶν μὴ

external arguments, whether of the existing is or of the non-existing,

ὄντων, εἶτα τίς ἡ φύσις αὐτοῦ. ὅτι μὲν οὖν ἢ ὅλως οὐκ

entities, then what the nature of it. that certainly therefore or entirely not

ἔστιν ἢ μόλις καὶ ἀμυδρῶς, ἐκ τῶνδέ τις ἂν ὑποπτεύσειεν.

is or hardly and faintly, from these someone might suspect.

τὸ μὲν γὰρ αὐτοῦ γέγονε καὶ οὐκ ἔστιν, τὸ δὲ μέλλει καὶ

the indeed for it has become and not is, the but will be and

οὔπω ἔστιν. [218a] ἐκ δὲ τούτων καὶ ὁ ἄπειρος καὶ ὁ ἀεὶ

not yet is. [218a] from these things and the infinite and the always

λαμβανόμενος χρόνος σύγκειται. τὸ δ' ἐκ μὴ ὄντων συγκείμενον

receiving time it is composed the and from not beings composed

ἀδύνατον ἂν εἶναι δόξειε μετέχειν οὐσίας.

impossible if to be it would seem to participate of substance

πρὸς δὲ τούτοις παντὸς μεριστοῦ, ἄνπερ ᾖ, ἀνάγκη, ὅτε

towards but these of all that which is divided, if it is, necessity, when

ἔστιν, ἤτοι πάντα τὰ μέρη εἶναι ἢ ἔνια· τοῦ δὲ χρόνου τὰ

it is, or all the parts to be or some things. of the but time the

μὲν γέγονε τὰ δὲ μέλλει, ἔστι δ' οὐδέν, ὄντος μεριστοῦ.

on one hand has occurred the but will happen, it is and nothing, being divided.

τὸ δὲ νῦν οὐ μέρος· μετρεῖ τε γὰρ τὸ μέρος, καὶ συγκεῖσθαι

the but now not part; measures and for the part, and to be composed

δεῖ τὸ ὅλον ἐκ τῶν μερῶν· ὁ δὲ χρόνος οὐ δοκεῖ
it is necessary the whole from the parts. the but time does not seem
συγκεῖσθαι ἐκ τῶν νῦν.
to be composed from the now.

ἔτι δὲ τὸ νῦν, ὃ φαίνεται διορίζειν τὸ παρελθὸν καὶ τὸ
furthermore but the now, which appears to distinguish the past and the
μέλλον, πότερον ἓν καὶ ταὐτὸν ἀεὶ διαμένει ἢ ἄλλο καὶ ἄλλο, οὐ
future, which one and the same always remains or another and another, not
ῥᾴδιον ἰδεῖν.
easy to see.

εἰ μὲν γὰρ αἰεὶ ἕτερον καὶ ἕτερον, μηδὲν δ' ἐστὶ τῶν ἐν
if on the one hand for always another and another, nothing but is of the in
τῷ χρόνῳ ἄλλο καὶ ἄλλο μέρος ἅμα (ὃ μὴ περιέχει, τὸ δὲ
the time another and another part together (which not contains, the but
περιέχεται, ὥσπερ ὁ ἐλάττων χρόνος ὑπὸ τοῦ πλείονος), τὸ δὲ
is contained, just as the lesser time by the greater), the but
νῦν μὴ ὂν πρότερον δὲ ὂν ἀνάγκη ἐφθάρθαι ποτέ, καὶ
now not being previously but being necessity to be destroyed at some time, and
τὰ νῦν ἅμα μὲν ἀλλήλοις οὐκ ἔσται, ἐφθάρθαι δὲ
the now together on the one hand to each other not will be, to be destroyed but
ἀνάγκη ἀεὶ τὸ πρότερον. ἐν αὑτῷ μὲν οὖν ἐφθάρθαι
necessity always the previously. in itself on the one hand therefore to be destroyed
οὐχ οἷόν τε διὰ τὸ εἶναι τότε, ἐν ἄλλῳ δὲ νῦν ἐφθάρθαι τὸ
not possible and through the being then, in another but now to be destroyed the
πρότερον νῦν οὐκ ἐνδέχεται. ἔστω γὰρ ἀδύνατον ἐχόμενα εἶναι
previously now not permitted. let it be for impossible conditions to be

ἀλλήλων τὰ νῦν, ὥσπερ στιγμὴν στιγμῆς. εἴπερ οὖν ἐν τῷ
of each other the now, just as a moment of time. if indeed therefore in the
ἐφεξῆς οὐκ ἔφθαρται ἀλλ' ἐν ἄλλῳ, ἐν τοῖς μεταξὺ [τοῖς] νῦν
subsequent not is corrupted but in another, in the midst [of] now
ἀπείροις οὖσιν ἅμα ἂν εἴη· τοῦτο δὲ ἀδύνατον.
infinite beings together if it should be. this but is impossible.

ἀλλὰ μὴν οὐδ' αἰεὶ τὸ αὐτὸ διαμένειν δυνατόν· οὐδενὸς
but on the other hand not even always the same to remain possible. of no one
γὰρ διαιρετοῦ πεπερασμένου ἓν πέρας ἔστιν, οὔτε ἂν ἐφ' ἓν ἦ
for separable limited one end exists, nor if on one is
συνεχὲς οὔτε ἂν ἐπὶ πλείω· τὸ δὲ νῦν πέρας ἐστίν, καὶ χρόνον
continuous nor if upon more. the but now end is, and time
ἔστι λαβεῖν πεπερασμένον. ἔτι εἰ τὸ ἅμα εἶναι κατὰ
is to take limited. moreover if the at the same time to be according to
χρόνον καὶ μήτε πρότερον μήτε ὕστερον τὸ ἐν τῷ αὐτῷ εἶναι καὶ
time and neither before nor after the in the same to be and
ἑνὶ [τῷ] νῦν ἐστιν, εἰ τά τε πρότερον καὶ τὰ ὕστερον ἐν τῷ νῦν
one [the] now is, if the and before and the after in the now
τῳδί ἐστιν, ἅμα ἂν εἴη τὰ ἔτος γενόμενα μυριοστὸν
this time is, at the same time if were the years having come to be myriad
τοῖς γε νομένοις τήμερον, καὶ οὔτε πρότερον οὔτε ὕστερον
to those indeed thinking today, and neither beforehand nor afterward
οὐδὲν ἄλλο ἄλλου.
nothing else of another.

περὶ μὲν οὖν τῶν ὑπαρχόντων αὐτῷ τοσαῦτ' ἔστω
concerning on one hand therefore of the existing to him so much let it be

διη πορημένα·

divided into passages;

τί δ' ἐστὶν ὁ χρόνος καὶ τίς αὐτοῦ ἡ φύσις, ὁμοίως ἔκ

what is it that is the time and what of it the nature, likewise from

τε τῶν παραδεδομένων ἄδηλόν ἐστιν, καὶ περὶ ὧν τυγχάνομεν

and of the handed down is unclear, it is, and concerning what we obtain

διεληλυθότες πρότερον.

having been analyzed beforehand.

οἱ μὲν γὰρ τὴν τοῦ ὅλου [218b] κίνησιν εἶναί φασιν, οἱ δὲ

those on one hand for the of the whole [218b] motion to be they say, those but

τὴν σφαῖραν αὐτήν.

the sphere itself.

καίτοι τῆς περιφορᾶς καὶ τὸ μέρος χρόνος τίς ἐστι, περιφορὰ δέ

yet of the revolution and the part time what is, the revolution but

γε οὔ· μέρος γὰρ περιφορᾶς τὸ ληφθέν, ἀλλ' οὐ περιφορά.

indeed is not. part for of the revolution the taken, but not revolution.

ἔτι δ' εἰ πλείους ἦσαν οἱ οὐρανοί, ὁμοίως ἂν ἦν ὁ χρόνος

further as for if more were the heavens, likewise would be the time

ἡ ὁτουοῦν αὐτῶν κίνησις, ὥστε πολλοὶ χρόνοι ἅμα.

the of whatever of them motion, so that many times together.

ἡ δὲ τοῦ ὅλου σφαῖρα ἔδοξε μὲν τοῖς εἰποῦσιν εἶναι ὁ χρόνος,

the but of the whole sphere seemed indeed to those saying to be the time,

ὅτι ἔν τε τῷ χρόνῳ πάντα ἐστὶν καὶ ἐν τῇ τοῦ ὅλου σφαίρᾳ·

that in and the time all is and in the of the whole sphere.

ἔστιν δ' εὐηθικώτερον τὸ εἰρημένον ἢ ὥστε περὶ αὐτοῦ τὰ

there is but more foolish the stated than so as to about it the

ἀδύνατα ἐπισκοπεῖν.

impossible to examine.

ἐπεὶ δὲ δοκεῖ μάλιστα κίνησις εἶναι καὶ μεταβολή τις ὁ χρόνος,

since but it seems especially motion to be and change some the time,

τοῦτ' ἂν εἴη σκεπτέον.

this would be to be considered.

ἡ μὲν οὖν ἑκάστου μεταβολὴ καὶ κίνησις ἐν αὐτῷ τῷ μεταβάλλοντι

the indeed therefore of each change and motion in it the changing

μόνον ἐστίν, ἢ οὗ ἂν τύχῃ ὂν αὐτὸ τὸ κινούμενον καὶ

only is, or of which at which it may happen being itself the moved and

μεταβάλλον· ὁ δὲ χρόνος ὁμοίως καὶ πανταχοῦ καὶ παρὰ πᾶσιν.

changing. the but time similarly and everywhere and beside all.

ἔτι δὲ μεταβολὴ μέν ἐστι θάττων καὶ βραδυτέρα, χρόνος δ'

yet but change on the one hand is quicker and slower, time but

οὐκ ἔστιν· τὸ γὰρ βραδὺ καὶ ταχὺ χρόνῳ ὥρισται, ταχὺ μὲν

not is. the for slow and fast in time has been defined, fast indeed

τὸ ἐν ὀλίγῳ πολὺ κινούμενον, βραδὺ δὲ τὸ ἐν πολλῷ ὀλίγον· ὁ

the in little much moving, slow but the in plenty little. the

δὲ χρόνος οὐχ ὥρισται χρόνῳ, οὔτε τῷ ποσός τις εἶναι

but time not has been defined in time, nor to the quantity someone to be

οὔτε τῷ ποιός.

nor to the quality.

ὅτι μὲν τοίνυν οὐκ ἔστιν κίνησις, φανερόν· μηδὲν δὲ διαφερέτω

that indeed therefore not is motion, clear; nothing but let differ

λέγειν ἡμῖν ἐν τῷ παρόντι κίνησιν ἢ μεταβολήν.

say to us in the present motion or change.

Κεφάλαιο 11

Chapter 11

Ἀλλὰ μὴν οὐδ' ἄνευ γε μεταβολῆς· ὅταν γὰρ μηδὲν
But indeed not even without at least change. Whenever for nothing
αὐτοὶ μεταβάλλωμεν τὴν διάνοιαν ἢ λάθωμεν μεταβάλλοντες, οὐ
they may change the mind or let us forget changing, not
δοκεῖ ἡμῖν γεγονέναι χρόνος, καθάπερ οὐδὲ τοῖς ἐν Σαρδοῖ
seems to us to have occurred time, just as not even to those in Sardis
μυθολογουμένοις καθεύδειν παρὰ τοῖς ἥρωσιν, ὅταν ἐγερθῶσι·
who are mythologizing to sleep near to the heroes, whenever they wake up;
συνάπτουσι γὰρ τῷ πρότερον νῦν τὸ ὕστερον νῦν καὶ ἓν ποιοῦσιν,
for they connect for the earlier now the later now and one they make,
ἐξαιροῦντες διὰ τὴν ἀναισθησίαν τὸ μεταξύ. ὥσπερ οὖν
taking away through the insensitivity the in between. Just as therefore
εἰ μὴ ἦν ἕτερον τὸ νῦν ἀλλὰ ταὐτὸ καὶ ἕν, οὐκ ἂν ἦν
if not it were another the now but the same and one, not would it be
χρόνος, οὕτως καὶ ἐπεὶ λανθάνει ἕτερον ὄν, οὐ δοκεῖ
time, in this way and since it escapes notice another existence, not it seems
εἶναι τὸ μεταξὺ χρόνος. εἰ δὴ τὸ μὴ οἴεσθαι εἶναι χρόνον τότε
to be the between time. If indeed the not to think to be time then
συμβαίνει ἡμῖν, ὅταν μὴ ὁρίσωμεν μηδεμίαν μεταβολήν, ἀλλ' ἐν ἑνὶ
occurs to us, when not we define any change, but in one
καὶ ἀδιαιρέτῳ φαίνηται ἡ ψυχὴ μένειν, ὅταν δ' αἰσθώμεθα καὶ
and indivisible it appears the soul to remain, when and we perceive and

ὁρίσωμεν, τότε φαμὲν γεγονέναι χρόνον, φανερὸν ὅτι οὐκ ἔστιν

we define, then we say to have come to be time, evident that not it is

ἄνευ κινήσεως καὶ μεταβολῆς [219a] χρόνος. ὅτι μὲν οὖν οὔτε

without motion and change [219a] time. that indeed therefore neither

κίνησις οὔτ' ἄνευ κινήσεως ὁ χρόνος ἐστί, φανερόν·

motion nor without motion the time is, evident;

ληπτέον δέ, ἐπεὶ ζητοῦμεν τί ἐστιν ὁ χρόνος,

it must be understood however, since we are seeking what is the time,

ἐντεῦθεν ἀρχομένοις, τί τῆς κινήσεώς ἐστιν.

thus beginning, what of the motion it is.

ἅμα γὰρ κινήσεως αἰσθανόμεθα καὶ χρόνου· καὶ γὰρ ἐὰν

at the same time for of movement we perceive and of time. and for if

ᾖ σκότος καὶ μηδὲν διὰ τοῦ σώματος πάσχωμεν, κίνησις

it is darkness and nothing through of the body we suffer, motion

δέ τις ἐν τῇ ψυχῇ ἐνῇ, εὐθὺς ἅμα δοκεῖ τις

but someone in the soul being, immediately at the same time it seems someone

γεγονέναι καὶ χρόνος. ἀλλὰ μὴν καὶ ὅταν γε χρόνος δοκῇ

to have happened and time. but indeed and when indeed time seems

γεγονέναι τις, ἅμα καὶ κίνησίς τις δοκεῖ γεγονέναι.

to have happened someone, at the same time and motion someone seems to have happened.

ὥστε ἤτοι κίνησις ἢ τῆς κινήσεώς τί ἐστιν ὁ χρόνος. ἐπεὶ

so that either motion or of the movement what is the time. since

οὖν οὐ κίνησις, ἀνάγκη τῆς κινήσεώς τι εἶναι αὐτόν.

therefore not motion, necessity of the movement something to be it.

ἐπεὶ δὲ τὸ κινούμενον κινεῖται ἔκ τινος εἴς τι καὶ πᾶν
since and the moving is moved from something to something and every
μέγεθος συνεχές, ἀκολουθεῖ τῷ μεγέθει ἡ κίνησις· διὰ γὰρ
magnitude continuous, follows the in magnitude the motion because of for
τὸ τὸ μέγεθος εἶναι συνεχὲς καὶ ἡ κίνησίς ἐστιν συνεχής, διὰ
the the magnitude to be continuous and the motion is continuous, because of
δὲ τὴν κίνησιν ὁ χρόνος· ὅση γὰρ ἡ κίνησις, τοσοῦτος καὶ
but the motion the time as much as for the motion, so much and
ὁ χρόνος αἰεὶ δοκεῖ γεγονέναι.
the time always seems to have been.

τὸ δὴ πρότερον καὶ ὕστερον ἐν τόπῳ πρῶτόν ἐστιν. ἐνταῦθα
the indeed earlier and later in place first is. here
μὲν δὴ τῇ θέσει· ἐπεὶ δ' ἐν τῷ μεγέθει ἔστι τὸ πρότερον
on one hand indeed the position: since but in the magnitude is the earlier
καὶ ὕστερον, ἀνάγκη καὶ ἐν κινήσει εἶναι τὸ πρότερον καὶ ὕστερον,
and later, necessity and in motion to be the earlier and later,
ἀνάλογον τοῖς ἐκεῖ. ἀλλὰ μὴν καὶ ἐν χρόνῳ ἔστιν τὸ πρότερον καὶ
proportional to the there. but indeed also in time is the earlier and
ὕστερον διὰ τὸ ἀκολουθεῖν ἀεὶ θατέρῳ θάτερον αὐτῶν. ἔστι
later through the following always one another the other of them. is
δὲ τὸ πρότερον καὶ ὕστερον ἐν τῇ κινήσει ὃ μέν ποτε
but the earlier and later in the motion which on one hand at some time
ὂν κίνησις [ἐστιν]· τὸ μέντοι εἶναι αὐτῷ ἕτερον καὶ οὐ κίνησις.
being motion [is]: the however to be to it another and not motion.
ἀλλὰ μὴν καὶ τὸν χρόνον γε γνωρίζομεν ὅταν ὁρίσωμεν τὴν
but indeed and the time for we know when we define the

κίνησιν, τῷ πρότερον καὶ ὕστερον ὁρίζοντες· καὶ τότε φαμὲν

motion, to the prior and later defining; and then we say

γεγονέναι χρόνον, ὅταν τοῦ προτέρου καὶ ὑστέρου ἐν τῇ κινήσει

to have become a time, when of the prior and later in the motion

αἴσθησιν λάβωμεν. ὁρίζομεν δὲ τῷ ἄλλο καὶ ἄλλο ὑπολαβεῖν

perception we receive. we define but to the other and another to suppose

αὐτά, καὶ μεταξύ τι αὐτῶν ἕτερον· ὅταν γὰρ ἕτερα τὰ ἄκρα

them, and between something of them another; when for other the extremes

τοῦ μέσου νοήσωμεν, καὶ δύο εἴπῃ ἡ ψυχὴ τὰ νῦν, τὸ μὲν

of the mean we conceive, and two would say the soul the now, the one

πρότερον τὸ δ' ὕστερον, τότε καὶ τοῦτό φαμεν εἶναι χρόνον· τὸ

prior the and later, then and this we say to be a time; the

γὰρ ὁριζόμενον τῷ νῦν χρόνος εἶναι δοκεῖ· καὶ ὑποκείσθω.

for defined by the now time to be it seems; and let it underlie.

ὅταν μὲν οὖν ὡς ἓν τὸ νῦν αἰσθανώμεθα, καὶ μὴ ἤτοι

whenever on the one hand therefore as one the now we perceive, and not either

ὡς πρότερον καὶ ὕστερον ἐν τῇ κινήσει ἢ ὡς τὸ αὐτὸ μὲν

as before and after in the motion or as the same on the one hand

προτέρου δὲ καὶ ὑστέρου τινός, οὐ δοκεῖ χρόνος γεγονέναι

of the former but and of the latter of some, not it seems time to have come into being

οὐδείς, ὅτι οὐδὲ κίνησις. ὅταν δὲ τὸ πρότερον [219b] καὶ

no one, because nor motion. whenever but the former [219b] and

ὕστερον, τότε λέγομεν χρόνον· τοῦτο γάρ ἐστιν ὁ χρόνος,

latter, then we say time, this for is the time,

ἀριθμὸς κινήσεως κατὰ τὸ πρότερον καὶ ὕστερον.

a number of motion according to the former and latter.

οὐκ ἄρα κί νησις ὁ χρόνος ἀλλ' ἢ ἀριθμὸν ἔχει ἡ κίνησις.
not therefore is island the time but that a number has the motion.
σημεῖον δέ· τὸ μὲν γὰρ πλεῖον καὶ ἔλαττον κρίνομεν
sign indeed; the on the one hand for more and less we judge
ἀριθμῷ, κίνησιν δὲ πλείω καὶ ἐλάττω χρόνῳ· ἀριθμὸς ἄρα τις
by number, motion but more and less time. number thus someone
ὁ χρόνος. ἐπεὶ δ' ἀριθμός ἐστι διχῶς (καὶ γὰρ τὸ ἀριθμούμενον
the time. since but number is in two ways (and for the counted
καὶ τὸ ἀριθμητὸν ἀριθμὸν λέγομεν, καὶ ᾧ ἀριθμοῦμεν), ὁ δὴ
and the counting number we say, and to which we count), the indeed
χρόνος ἐστὶν τὸ ἀριθμούμενον καὶ οὐχ ᾧ ἀριθμοῦμεν. ἔστι
time is the counted and not to which we count. there is
δ' ἕτερον ᾧ ἀριθμοῦμεν καὶ τὸ ἀριθμούμενον. καὶ ὥσπερ ἡ
but another to which we count and the counted. And just as the
κίνησις αἰεὶ ἄλλη καὶ ἄλλη, καὶ ὁ χρόνος (ὁ δ' ἅμα πᾶς
motion always another and another, and the time (the but at once all
χρόνος ὁ αὐτός· τὸ γὰρ νῦν τὸ αὐτὸ ὅ ποτ' ἦν-τὸ δ' εἶναι αὐτῷ
time the same; for indeed now the same as ever it was- but to be for it
ἕτερον-τὸ δὲ νῦν τὸν χρόνον ὁρίζει, ᾗ πρότερον καὶ ὕστερον).
another- but now the time defines, when before and after

τὸ δὲ νῦν ἔστι μὲν ὡς τὸ αὐτό, ἔστι δ' ὡς οὐ τὸ αὐτό· ᾗ
the and now is on one hand as the same, it is but as not the same. in which
μὲν γὰρ ἐν ἄλλῳ καὶ ἄλλῳ, ἕτερον (τοῦτο δ' ἦν αὐτῷ τὸ
on one hand for in another and another, different (this but was to it the
νῦν <εἶναι>, ὃ δέ ποτε ὄν ἐστι τὸ νῦν, τὸ αὐτό. ἀκολουθεῖ
now <to be>, which and once being is the now, the same. follows

γάρ, ὡς ἐλέχθη, τῷ μὲν μεγέθει ἡ κίνησις, ταύτῃ δ' ὁ
for, as it was said, to the on one hand magnitude the motion, in this but the
χρόνος, ὥς φαμεν· καὶ ὁμοίως δὴ τῇ στιγμῇ τὸ φερόμενον,
time, as we say; and similarly indeed to the moment the carried,
ᾧ τὴν κίνησιν γνωρίζομεν καὶ τὸ πρότερον ἐν αὐτῇ καὶ τὸ
to which the motion we recognize and the earlier in it and the
ὕστερον. τοῦτο δὲ ὃ μέν ποτε ὂν τὸ αὐτό (ἡ στιγμὴ γὰρ
later. this and which on one hand once being the same (or moment for
ἡ λίθος ἤ τι ἄλλο τοιοῦτόν ἐστι), τῷ λόγῳ δὲ ἄλλο, ὥσπερ
or stone or something else such is) to the word but another, just as
οἱ σοφισταὶ λαμβάνουσιν ἕτερον τὸ Κορίσκον ἐν Λυκείῳ εἶναι καὶ
the sophists take another the Koriskon in Lyceum to be and
τὸ Κορίσκον ἐν ἀγορᾷ. καὶ τοῦτο δὴ τῷ ἄλλοθι καὶ ἄλλοθι
the Koriskon in market. and this indeed to the elsewhere and elsewhere
εἶναι ἕτερον· τῷ δὲ φερομένῳ ἀκολουθεῖ τὸ νῦν, ὥσπερ ὁ
is another· to the but carried follows the now, just as the
χρόνος τῇ κινήσει (τῷ γὰρ φερομένῳ γνωρίζομεν τὸ πρότερον καὶ
time in the motion (the for carried we know the former and
ὕστερον ἐν κινήσει, ᾗ δ' ἀριθμητὸν τὸ πρότερον καὶ ὕστερον,
latter in motion, which that is measurable the former and latter,
τὸ νῦν ἔστιν)· ὥστε καὶ ἐν τούτοις ὃ μέν ποτε ὂν νῦν
the now is).· so and in these which indeed at some time being now
ἐστι, τὸ αὐτό (τὸ πρότερον γὰρ καὶ ὕστερόν ἐστι τὸ ἐν κινήσει),
is, the same (the former for and latter is the in motion),
τὸ δ' εἶναι ἕτερον (ᾗ ἀριθμητὸν γὰρ τὸ πρότερον καὶ ὕστερον,
the but to be another (which can be counted for the former and latter,
τὸ νῦν ἔστιν). καὶ γνώριμον δὲ μάλιστα τοῦτ' ἔστιν· καὶ γὰρ ἡ
the now exists). And recognizable but particularly this is; And for the

κίνησις διὰ τὸ κινούμενον καὶ ἡ φορὰ διὰ τὸ φερόμενον·

motion through the moving and the carrying through the carried;

τόδε γάρ τι τὸ φερόμενον, ἡ δὲ κίνησις οὔ. ἔστι μὲν

this for something the carried, the but motion is not. It is indeed

οὖν ὡς τὸ αὐτὸ τὸ νῦν αἰεί, ἔστι δ' ὡς οὐ τὸ αὐτό· καὶ γὰρ

therefore as the same the now always, it is but as not the same; and for

τὸ φερόμενον. φανερὸν δὲ καὶ ὅτι εἴτε χρόνος μὴ εἴη, τὸ νῦν οὐκ

the carried. Evident but also that whether time not be, the now not

ἂν [220a] εἴη, εἴτε τὸ νῦν μὴ εἴη, χρόνος οὐκ ἂν εἴη· ἅμα

would [220a] be, whether the now not be, time not would be· together

γὰρ ὥσπερ τὸ φερόμενον καὶ ἡ φορά, οὕτως καὶ ὁ ἀριθμὸς ὁ

for just as the carried and the motion, so and the number the

τοῦ φερομένου καὶ ὁ τῆς φορᾶς.

of the carried and the of the motion.

χρόνος μὲν γὰρ ὁ τῆς φορᾶς ἀριθμός, τὸ νῦν δὲ ὡς τὸ

time on one hand for the of the carrying number, the now but as the

φερόμενον, οἷον μονὰς ἀριθμοῦ. καὶ συνεχής τε δὴ ὁ χρόνος

carried, such as unit of number. and continuous and indeed the time

τῷ νῦν, καὶ διῄρηται κατὰ τὸ νῦν· ἀκολουθεῖ γὰρ καὶ τοῦτο

to the now, and is divided according to the now; it follows for and this

τῇ φορᾷ καὶ τῷ φερομένῳ. καὶ γὰρ ἡ κίνησις καὶ ἡ φορὰ

to the carrying and to the carried. and for the motion and the carrying

μία τῷ φερομένῳ, ὅτι ἕν (καὶ οὐχ ὅ ποτε ὄν-καὶ γὰρ ἂν

one to the carried, because one (and not that ever being - and for if

διαλίποι-ἀλλὰ τῷ λόγῳ)· καὶ ὁρίζει δὲ τὴν πρότερον

it were to separate - but to the reason); and it defines but the earlier

καὶ ὕστερον κίνησιν τοῦτο. ἀκολουθεῖ δὲ καὶ τοῦτό πως τῇ

and later motion this. it follows but and this somehow to the

στιγμῇ· καὶ γὰρ ἡ στιγμὴ καὶ συνέχει τὸ μῆκος καὶ ὁρίζει· ἔστι

moment; and for the moment and continues the length and defines. it is

γὰρ τοῦ μὲν ἀρχὴ τοῦ δὲ τελευτή. ἀλλ' ὅταν μὲν οὕτω

for of the one beginning of the other end. But when one in this way

λαμβάνῃ τις ὡς δυσὶ χρώμενος τῇ μιᾷ, ἀνάγκη ἵστασθαι, εἰ

takes someone as two using the one, necessity to stand, if

ἔσται ἀρχὴ καὶ τελευτὴ ἡ αὐτὴ στιγμή· τὸ δὲ νῦν διὰ τὸ

there will be a beginning and an end the same moment. The other now by the

κινεῖσθαι τὸ φερόμενον αἰεὶ ἕτερον.

movement the carried always other.

ὥσθ' ὁ χρόνος ἀριθμὸς οὐχ ὡς τῆς αὐτῆς στιγμῆς, ὅτι

Thus the time number is not as that of the same moment, since

ἀρχὴ καὶ τελευτή, ἀλλ' ὡς τὰ ἔσχατα τῆς γραμμῆς μᾶλλον-καὶ

beginning and end, but as the extremes of the line rather-and

οὐχ ὡς τὰ μέρη, διά τε τὸ εἰρημένον (τῇ γὰρ μέσῃ στιγμῇ ὡς

not as the parts, through and the aforementioned (to the for middle moment as

δυσὶ χρήσεται, ὥστε ἠρεμεῖν συμβήσεται), καὶ ἔτι φανερὸν ὅτι

two will be used, so that to be at rest will happen) and yet evident that

οὐδὲν μόριον τὸ νῦν τοῦ χρόνου, οὐδ' ἡ διαίρεσις τῆς κινήσεως,

nothing part the now of the time, nor the division of the motion,

ὥσπερ οὐδ' ἡ στιγμὴ τῆς γραμμῆς· αἱ δὲ γραμμαὶ αἱ δύο τῆς

just as nor the moment of the line. the but letters the two of the

μιᾶς μόρια.

one parts.

ᾗ μὲν οὖν πέρας τὸ νῦν, οὐ χρόνος, ἀλλὰ συμβέβηκεν·

in which on the one hand therefore limit the now, not time, but has occurred.

ᾗ δ' ἀριθμεῖ, ἀριθμός· τὰ μὲν γὰρ πέρατα ἐκείνου

in which and counts, number . the on the one hand for limits of that

μόνον ἐστὶν οὗ ἐστιν πέρατα, ὁ δ' ἀριθμὸς ὁ τῶνδε τῶν

only is of which it is limits, the and number the of these the

ἵππων, ἡ δεκάς, καὶ ἄλλοθι.

horses, the decade, and elsewhere.

ὅτι μὲν τοίνυν ὁ χρόνος ἀριθμός ἐστιν κινήσεως κατὰ

that on the one hand therefore the time is a number is of motion according to

τὸ πρότερον καὶ ὕστερον, καὶ συνεχής (συνεχοῦς γάρ), φανερόν.

the earlier and later, and continuous (continuous for), it is evident.

Κεφάλαιο 12

Chapter 12

Ἐλάχιστος δὲ ἀριθμὸς ὁ μὲν ἁπλῶς ἐστὶν ἡ δυάς· τὶς

The least but number the on the one hand simply is the pair. someone

δὲ ἀριθμὸς ἔστι μὲν ὡς ἔστιν, ἔστι δ' ὡς οὐκ ἔστιν, οἷον

but number is on the one hand as it is, is and as not it is, such as

γραμμῆς ἐλάχιστος πλήθει μέν ἐστιν αἱ δύο ἢ ἡ μία,

of a line least in quantity on the one hand it is the two or the one,

μεγέθει δ' οὐκ ἔστιν ἐλάχιστος· ἀεὶ γὰρ διαιρεῖται πᾶσα γραμμή.

in size but not is the least. always for is divided every line.

ὥστε ὁμοίως καὶ χρόνος· ἐλάχιστος γὰρ κατὰ μὲν ἀριθμόν

so likewise and time. the least for in on one hand number

ἐστιν ὁ εἷς ἢ οἱ δύο, κατὰ μέγεθος δ' οὐκ ἔστιν.

is the one or the two, in terms of size but not is.

φανερὸν [220b] δὲ καὶ ὅτι ταχὺς μὲν καὶ βραδὺς οὐ λέγεται,

evident [220b] but also that swift on one hand and slow not said,

πολὺς δὲ καὶ ὀλίγος καὶ μακρὸς καὶ βραχύς. ᾗ μὲν γὰρ

much but and little and long and short. in which on one hand for

συνεχής, μακρὸς καὶ βραχύς, ᾗ δὲ ἀριθμός, πολὺς καὶ ὀλίγος.

continuous, long and short, in which but number, many and few.

ταχὺς δὲ καὶ βραδὺς οὐκ ἔστιν· οὐδὲ γὰρ ἀριθμὸς ᾗ ἀριθμοῦμεν

swift but and slow not is. nor for number in which we count

ταχὺς καὶ βραδὺς οὐδείς. καὶ ὁ αὐτὸς δὲ πανταχοῦ ἅμα· πρότερον

swift and slow none. and the same but everywhere at once; before

δὲ καὶ ὕστερον οὐχ ὁ αὐτός, ὅτι καὶ ἡ μεταβολὴ ἡ μὲν

but and after not the same, because and the change the on one hand

παροῦσα μία, ἡ δὲ γεγενημένη καὶ ἡ μέλλουσα ἑτέρα, ὁ δὲ

present one, the but having come to be and the future another, the but

χρόνος ἀριθμός ἐστιν οὐχ ᾧ ἀριθμοῦμεν ἀλλ' ὁ ἀριθμούμενος,

time is a number is not to which we count but the counted,

οὗτος δὲ συμβαίνει πρότερον καὶ ὕστερον ἀεὶ ἕτερος· τὰ γὰρ νῦν

this but happens before and after always another; the for now

ἕτερα. ἔστι δὲ ὁ ἀριθμὸς εἷς μὲν καὶ ὁ αὐτὸς ὁ τῶν

different. there is but the number one on one hand and the same the of the

ἑκατὸν ἵππων καὶ ὁ τῶν ἑκατὸν ἀνθρώπων, ὧν δ' ἀριθμός,

hundred horses and the of the hundred men, of whom and the number,

ἕτερα, οἱ ἵπποι τῶν ἀνθρώπων. ἔτι ὡς ἐνδέχεται κίνησιν εἶναι τὴν

other, the horses of the men. also as it is possible movement to be the

αὐτὴν καὶ μίαν πάλιν καὶ πάλιν, οὕτω καὶ χρόνον, οἷον ἐνιαυτὸν ἢ

same and one again and again, thus and time, such as a year or

ἔαρ ἢ μετόπωρον.

spring or autumn.

οὐ μόνον δὲ τὴν κίνησιν τῷ χρόνῳ μετροῦμεν, ἀλλὰ καὶ τῇ κινήσει

not only but the motion with time we measure, but and the motion

τὸν χρόνον διὰ τὸ ὁρίζεσθαι ὑπ' ἀλλήλων· ὁ μὲν γὰρ

the time through the to be defined by each other. the on one hand for

χρόνος ὁρίζει τὴν κίνησιν ἀριθμὸς ὢν αὐτῆς, ἡ δὲ

time defines the motion number being of it, the on the other hand

κίνησις τὸν χρόνον. καὶ λέγομεν πολὺν καὶ ὀλίγον χρόνον τῇ

motion the time. And we say much and little time the

κινήσει μετροῦντες, καθάπερ καὶ τῷ ἀριθμητῷ τὸν ἀριθμόν, οἷον

motion measuring, just as and the number the number, such as

τῷ ἑνὶ ἵππῳ τὸν τῶν ἵππων ἀριθμόν. τῷ μὲν γὰρ ἀριθμῷ

the one horse the of the horses number. the on one hand for number

τὸ τῶν ἵππων πλῆθος γνωρίζομεν, πάλιν δὲ τῷ ἑνὶ ἵππῳ τὸν

the of the horses quantity we recognize, again but the one horse the

τῶν ἵππων ἀριθμὸν αὐτόν. ὁμοίως δὲ καὶ ἐπὶ τοῦ χρόνου

of the horses number itself. Likewise but and regarding the of time

καὶ τῆς κινήσεως· τῷ μὲν γὰρ χρόνῳ τὴν κίνησιν, τῇ

and of the motion· to the on one hand for time the motion, to the

δὲ κινήσει τὸν χρόνον μετροῦμεν. καὶ τοῦτ' εὐλόγως

on the other hand motion the time we measure. And this understandably

συμβέβηκεν· ἀκολουθεῖ γὰρ τῷ μὲν μεγέθει ἡ κίνησις, τῇ

has happened· it follows for to the on one hand magnitude the motion, to the

δὲ κινήσει ὁ χρό νος, τῷ καὶ ποσὰ καὶ συνεχῆ καὶ διαιρετὰ

but motion the time - to the and quantity and continuous and divisible

εἶναι· διὰ μὲν γὰρ τὸ τὸ μέγεθος εἶναι τοιοῦτον ἡ κίνησις

to be· through on one hand for the the magnitude to be such the motion

ταῦτα πέπονθεν, διὰ δὲ τὴν κίνησιν ὁ χρόνος. καὶ μετροῦμεν

these has undergone, through but the motion the time. And we measure

καὶ τὸ μέγεθος τῇ. κινήσει καὶ τὴν κίνησιν τῷ μεγέθει·

and the magnitude to the. motion and the motion to the magnitude·

πολλὴν γὰρ εἶναί φαμεν τὴν ὁδόν, ἂν ἡ πορεία πολλή, καὶ ταύτην

much for to be we say the way, if the progression much, and this

πολλήν, ἂν ἡ ὁδὸς [ἥ] πολλή· καὶ τὸν χρόνον, ἂν ἡ κίνησις,

much, if the road [may be] much. And the time, if the motion,

καὶ τὴν κίνησιν, ἂν ὁ χρόνος.

and the motion, if the time.

ἐπεὶ δ' ἐστὶν ὁ χρόνος μέτρον [221a] κινήσεως καὶ τοῦ κινεῖσθαι,

since but it is the time measure [221a] of motion and of the to be moved,

μετρεῖ δ' οὗτος τὴν κίνησιν τῷ ὁρίσαι τινὰ κίνησιν ἣ

it measures but this the motion to the define some motion which

καταμετρήσει τὴν ὅλην (ὥσπερ καὶ τὸ μῆκος ὁ πῆχυς τῷ ὁρίσαι

will measure the entire (just as and the length the cubit to the define

τι μέγεθος ὃ ἀναμετρήσει τὸ ὅλον), καὶ ἔστιν τῇ κινήσει τὸ

some magnitude which will measure the whole), and it is to the motion the

ἐν χρόνῳ εἶναι τὸ μετρεῖσθαι τῷ χρόνῳ καὶ αὐτὴν καὶ τὸ εἶναι

in time to be the to be measured to the time and itself and the to be

αὐτῆς (ἅμα γὰρ τὴν κίνησιν καὶ τὸ εἶναι τῆς κινήσεως

of it (at the same time for the motion and the to be of the motion

μετρεῖ, καὶ τοῦτ' ἔστιν αὐτῇ τὸ ἐν χρόνῳ εἶναι, τὸ μετρεῖσθαι

measures, and this is in it the in time to be, the to be measured

αὐτῆς τὸ εἶναι), δῆλον ὅτι καὶ τοῖς ἄλλοις τοῦτ' ἔστι τὸ ἐν
of it the to be), it is clear that and to the others this is the in
χρόνῳ εἶναι, τὸ μετρεῖσθαι αὐτῶν τὸ εἶναι ὑπὸ τοῦ χρόνου. τὸ
time to be, the to be measured of them the to be by the time. the
γὰρ ἐν χρόνῳ εἶναι δυοῖν ἐστιν θάτερον, ἓν μὲν τὸ εἶναι
for in time to be of two is the other, one on the one hand the to be
τότε ὅτε ὁ χρόνος ἔστιν, ἓν δὲ ὥσπερ ἔνια λέγομεν ὅτι
then when the time is, one on the other hand just as some we say that
ἐν ἀριθμῷ ἐστιν. τοῦτο δὲ σημαίνει ἤτοι ὡς μέρος ἀριθμοῦ καὶ
in number exists. this but means either as a part of a number and
πάθος, καὶ ὅλως ὅτι τοῦ ἀριθμοῦ τι, ἢ ὅτι ἔστιν αὐτοῦ
a passion, and altogether that of the number something, or that it exists of it
ἀριθμός.
as a number.

ἐπεὶ δ' ἀριθμὸς ὁ χρόνος, τὸ μὲν νῦν καὶ τὸ πρότερον καὶ ὅσα
since but number the time, the on one hand now and the earlier and all
τοιαῦτα οὕτως ἐν χρόνῳ ὡς ἐν ἀριθμῷ μονὰς καὶ τὸ περιττὸν καὶ
such thus in time as in number units and the odd and
ἄρτιον (τὰ μὲν γὰρ τοῦ ἀριθμοῦ τι, τὰ δὲ τοῦ χρόνου
even (the on one hand for of the number something, the but of the time
τί ἐστιν)· τὰ δὲ πράγματα ὡς ἐν ἀριθμῷ τῷ χρόνῳ ἐστίν. εἰ
what is)· the but things as in number the time is. if
δὲ τοῦτο, περιέχεται ὑπὸ χρόνου ὥσπερ <καὶ τὰ ἐν ἀριθμῷ ὑπ'
but this, is contained by time just as <and the in number by
ἀριθμοῦ> καὶ τὰ ἐν τόπῳ ὑπὸ τόπου.
number> and the in place by place.

φανερὸν δὲ καὶ ὅτι οὐκ ἔστιν τὸ ἐν χρόνῳ εἶναι τὸ εἶναι ὅτε ὁ

evident but also that not is the in time to be the to be when the

χρόνος ἔστιν, ὥσπερ οὐδὲ τὸ ἐν κινήσει εἶναι οὐδὲ τὸ ἐν τόπῳ

time is, just as neither the in motion to be nor the in place

ὅτε ἡ κίνησις καὶ ὁ τόπος ἔστιν. εἰ γὰρ ἔσται τὸ ἔν τινι

when the motion and the place is. if for it will be the in something

οὕτω, πάντα τὰ πράγματα ἐν ὁτῳοῦν ἔσται, καὶ ὁ οὐρανὸς ἐν τῇ

thus, all the things in any will be, and the heaven in the

κέγχρῳ· ὅτε γὰρ ἡ κέγχρος ἔστιν, ἔστι καὶ ὁ οὐρανός. ἀλλὰ

chaos; when for the chaos is, it is and the heaven. But

τοῦτο μὲν συμβέβηκεν, ἐκεῖνο δ' ἀνάγκη παρακολουθεῖν, καὶ τῷ

this indeed has happened, that but necessity to follow, and to the

ὄντι ἐν χρόνῳ εἶναί τινα χρόνον ὅτε κἀκεῖνο ἔστιν, καὶ τῷ ἐν

actual in time to be some time when and that is, and to the in

κινήσει ὄντι εἶναι τότε κίνησιν.

motion actual to be then motion.

ἐπεὶ δέ ἐστιν ὡς ἐν ἀριθμῷ τὸ ἐν χρόνῳ, ληφθήσεταί τις πλείων

since but it is as in number the in time, will be taken some greater

χρόνος παντὸς τοῦ ἐν χρόνῳ ὄντος· διὸ ἀνάγκη πάντα τὰ ἐν

time of all the in time being. Therefore necessity all the in

χρόνῳ ὄντα περιέχεσθαι ὑπὸ χρόνου, ὥσπερ καὶ τἆλλα ὅσα

time being to be contained by time, just as and the others as many as

ἔν τινί ἐστιν, οἷον τὰ ἐν τόπῳ ὑπὸ τοῦ τόπου.

are in a certain place, such as the in place by the place.

καὶ πάσχει δή τι ὑπὸ τοῦ χρόνου, καθάπερ καὶ λέγειν

and suffers indeed something under the time, just as and to say

εἰώθαμεν ὅτι κατατήκει ὁ χρόνος, καὶ γηράσκει πάνθ'
we are accustomed that it is wasting away the time, and grows old all things
ὑπὸ τοῦ χρόνου, καὶ ἐπιλανθάνεται διὰ τὸν χρόνον, ἀλλ' [221b]
under the time, and forgets through the time, but [221b]
οὐ μεμάθηκεν, οὐδὲ νέον γέγονεν οὐδὲ καλόν· φθορᾶς γὰρ αἴτιος
not has learned, nor new has become nor good. of decay for the cause
καθ' ἑαυτὸν μᾶλλον ὁ χρόνος· ἀριθμὸς γὰρ κινήσεως, ἡ δὲ
for itself mostly the time. number for of motion, the but
κίνησις ἐξίστησιν τὸ ὑπάρχον· ὥστε φανερὸν ὅτι τὰ αἰεὶ
motion causes to stand out the existing; so evident that the always
ὄντα, ᾗ αἰεὶ ὄντα, οὐκ ἔστιν ἐν χρόνῳ· οὐ γὰρ περιέχεται
existing, in which always existing, not is in time. not for is contained
ὑπὸ χρόνου, οὐδὲ μετρεῖται τὸ εἶναι αὐτῶν ὑπὸ τοῦ χρόνου·
under time, nor is measured the being of them under the time.
σημεῖον δὲ τούτου ὅτι οὐδὲ πάσχει οὐδὲν ὑπὸ τοῦ χρόνου ὡς οὐκ
a sign but of this that nor suffers nothing by the time as not
ὄντα ἐν χρόνῳ. ἐπεὶ δ' ἐστὶν ὁ χρόνος μέτρον κινήσεως, ἔσται
being in time. since but it is the time measure of motion, it will be
καὶ ἠρεμίας μέτρον [κατὰ συμβεβηκός]· πᾶσα γὰρ ἠρεμία
also of rest measure according to accident]· every for rest
ἐν χρόνῳ. οὐ γὰρ ὥσπερ τὸ ἐν κινήσει ὂν ἀνάγκη κινεῖσθαι,
in time. not for just as the in motion being necessity to be moved,
οὕτω καὶ τὸ ἐν χρόνῳ· οὐ γὰρ κίνησις ὁ χρόνος, ἀλλ' ἀριθμὸς
so also the in time· not for motion the time, but a number
κινήσεως, ἐν ἀριθμῷ δὲ κινήσεως ἐνδέχεται εἶναι καὶ τὸ ἠρεμοῦν.
of motion, in number but of motion may happen to be and the resting.
οὐ γὰρ πᾶν τὸ ἀκίνητον ἠρεμεῖ, ἀλλὰ τὸ ἐστερημένον κινήσεως
not for all the immovable rests, but the deprived of motion

πεφυκὸς δὲ κινεῖσθαι, καθάπερ εἴρηται ἐν τοῖς πρότερον.

by nature and to be moved, just as has been said in the earlier.

τὸ δ' εἶναι ἐν ἀριθμῷ ἐστιν τὸ εἶναί τινα ἀριθμὸν τοῦ πράγματος,

the but to be in number is the to be some number of the thing,

καὶ μετρεῖσθαι τὸ εἶναι αὐτοῦ τῷ ἀριθμῷ ἐν ᾧ ἐστιν, ὥστ' εἰ

and to be measured the to be of it to the number in which is, so that if

ἐν χρόνῳ, ὑπὸ χρόνου. μετρήσει δ' ὁ χρόνος τὸ κινούμενον

in time, by time. it will measure but the time the moving

καὶ τὸ ἠρεμοῦν, ᾗ τὸ μὲν κινούμενον τὸ δὲ

and the at rest, in which the on the one hand moving the on the other hand

ἠρεμοῦν· τὴν γὰρ κίνησιν αὐτῶν μετρήσει καὶ τὴν ἠρεμίαν,

resting; the for movement of them it will measure and the rest,

πόση τις.

how much some.

ὥστε τὸ κινούμενον οὐχ ἁπλῶς ἔσται μετρητὸν ὑπὸ χρόνου, ᾗ

so that the moving not simply will be measurable by time, in which

ποσόν τί ἐστιν, ἀλλ' ᾗ ἡ κίνησις αὐτοῦ ποσή. ὥστε ὅσα

quantity what it is, but in which the movement of it how much. so that as many as

μήτε κινεῖται μήτ' ἠρεμεῖ, οὐκ ἔστιν ἐν χρόνῳ· τὸ μὲν γὰρ ἐν

neither moves nor rests, not it is in time. the on one hand for in

χρόνῳ εἶναι τὸ μετρεῖσθαί ἐστι χρόνῳ, ὁ δὲ χρόνος κινήσεως καὶ

time to be the to be measured is in time, the but time of motion and

ἠρεμίας μέτρον.

of rest is a measure.

φανερὸν οὖν ὅτι οὐδὲ τὸ μὴ ὂν ἔσται πᾶν ἐν χρόνῳ, οἷον

evident therefore that neither the not being will be all in time, such as

ὅσα μὴ ἐνδέχεται ἄλλως, ὥσπερ τὸ τὴν διάμετρον εἶναι τῇ
whatever not might be possible otherwise, just as the the diameter to be to the
πλευρᾷ σύμμετρον.
side is in proportion.

ὅλως γάρ, εἰ μέτρον μέν ἐστι κινήσεως ὁ χρόνος καθ'
entirely for, if a measure on one hand is of motion the time according to
αὐτό, τῶν δ' ἄλλων κατὰ συμβεβηκός, δῆλον ὅτι
itself, of the but others in accordance with what has happened, it is clear that
ὧν τὸ εἶναι μετρεῖ, τούτοις ἅπασιν ἔσται τὸ εἶναι ἐν τῷ ἠρεμεῖν
of what the being measures, with these all will be the being in the resting
ἢ κινεῖσθαι. ὅσα μὲν οὖν φθαρτὰ καὶ γενητὰ καὶ ὅλως
or being moved. As many as indeed thus perishable and generated and entirely
ὁτὲ μὲν ὄντα ὁτὲ δὲ μή, ἀνάγκη ἐν χρόνῳ εἶναι (ἔστιν
sometimes indeed being sometimes but not, necessity in time to be (there is
γὰρ χρόνος τις πλείων, ὃς ὑπερέξει τοῦ τε εἶναι αὐτῶν καὶ
for time some greater, which will surpass the and being of them and
τοῦ μετροῦντος τὴν οὐσίαν αὐτῶν)· τῶν δὲ μὴ ὄντων ὅσα
the measuring the essence of them); of those but not being as many as
μὲν περιέχει ὁ χρόνος, τὰ μὲν ἦν, οἷον Ὅμηρός ποτε [222a]
indeed encompasses the time, the indeed was, such as Homer ever [222a]
ἦν, τὰ δὲ ἔσται, οἷον τῶν μελλόντων τι, ἐφ' ὁπότερα
was, the but will be, such as of the future something, regarding whichever
περιέχει· καὶ εἰ ἐπ' ἄμφω, ἀμφότερα [καὶ ἦν καὶ ἔσται]· ὅσα
it contains; and if on both, both and was and will be]; as many as
δὲ μὴ περιέχει μηδαμῇ, οὔτε ἦν οὔτε ἔστιν οὔτε ἔσται. ἔστι δὲ τὰ
but not it contains in no way, neither was nor is nor will be. it is but the

τοιαῦτα τῶν μὴ ὄντων, ὅσων τἀντικείμενα ἀεὶ ἔστιν, οἷον τὸ

such things of the not existing, of which the opposites always exist, such as the

ἀσύμμετρον εἶναι τὴν διάμετρον ἀεὶ ἔστι, καὶ οὐκ ἔσται τοῦτ' ἐν

asymmetrical to be the diameter always is, and not will be this in

χρόνῳ. οὐ τοίνυν οὐδὲ τὸ σύμμετρον· διὸ ἀεὶ οὐκ ἔστιν,

time. not therefore not even the symmetrical; therefore always not is,

ὅτι ἐναντίον τῷ ἀεὶ ὄντι. ὅσων δὲ τὸ ἐναντίον μὴ ἀεί,

because opposite to the always existing. as many as but the opposite not always,

ταῦτα δὲ δύναται καὶ εἶναι καὶ μή, καὶ ἔστιν γένεσις καὶ φθορὰ

these but can and to be and not, and there is origin and decay

αὐτῶν.

of them.

Κεφάλαιο 13

Chapter 13

Τὸ δὲ νῦν ἐστιν συνέχεια χρόνου, ὥσπερ ἐλέχθη· συνέχει γὰρ τὸν

The but now is continuity of time, just as was said; it continues for the

χρόνον τὸν παρεληλυθότα καὶ ἐσόμενον, καὶ πέρας χρόνου ἐστίν·

time the having passed and will be, and the limit of time is;

ἔστι γὰρ τοῦ μὲν ἀρχή, τοῦ δὲ τελευτή. ἀλλὰ τοῦτ' οὐχ

it is for of indeed the beginning, of but the end. But this is not

ὥσπερ ἐπὶ τῆς στιγμῆς μενούσης φανερόν. διαιρεῖ δὲ δυνάμει. καὶ

as at the moment remaining clear. It divides and by power. And

ᾗ μὲν τοιοῦτο, ἀεὶ ἕτερον τὸ νῦν, ᾗ δὲ συνδεῖ, ἀεὶ τὸ

in which indeed such, always another the now, in which but connects, always the

αὐτό, ὥσπερ ἐπὶ τῶν μαθηματικῶν γραμμῶν (οὐ γὰρ ἡ αὐτὴ ἀεὶ

same, just as at the mathematical lines (not for the same always

στιγμὴ τῇ νοήσει· διαιρούντων γὰρ ἄλλη καὶ ἄλλη· ᾗ δὲ

moment to the thought; of those dividing for another and another; in which but

μία, ἡ αὐτὴ πάντῃ)

one, the same everywhere)

-οὕτω καὶ τὸ νῦν τὸ μὲν τοῦ χρόνου διαίρεσις κατὰ

-thus and the now the on one hand of the time division according to

δύναμιν, τὸ δὲ πέρας ἀμφοῖν καὶ ἑνότης· ἔστι δὲ ταὐτὸ καὶ

power, the but limit of both and unity. there is but the same also

κατὰ ταὐτὸ ἡ διαίρεσις καὶ ἡ ἕνωσις, τὸ δ' εἶναι οὐ ταὐτό.

according to the same the division and the union, the but to be not the same.

τὸ μὲν οὖν οὕτω λέγεται τῶν νῦν, ἄλλο δ' ὅταν ὁ

the on one hand therefore thus it is said of those now, another but when the

χρόνος ὁ τούτου ἐγγὺς ᾖ. ἥξει νῦν, ὅτι τήμερον

time the of this near may be. it will come now, because today

ἥξει· ἧκει νῦν, ὅτι ἦλθε τήμερον. τὰ δ' ἐν Ἰλίῳ

it will come. it has come now, because it came today. the but in Ilium

γέγονεν οὐ νῦν, οὐδ' ὁ κατακλυσμὸς [γέγονε] νῦν· καίτοι

has happened not now, nor the flood [has happened] now. and yet

συνεχὴς ὁ χρόνος εἰς αὐτά, ἀλλ' ὅτι οὐκ ἐγγύς.

continuous the time into these, but because not near.

τὸ δὲ ποτέ χρόνος ὡρισμένος πρὸς τὸ πρότερον νῦν, οἷον

the and at any time time defined in relation to the previous now, such as

ποτὲ ἐλήφθη Τροία, καὶ ποτὲ ἔσται κατακλυσμός· δεῖ γὰρ

once was taken Troy, and once will be a flood. it is necessary for

πεπεράνθαι πρὸς τὸ νῦν. ἔσται ἄρα ποσός τις ἀπὸ τοῦδε
to be completed in relation to the now. it will be thus how much someone from this
χρόνος εἰς ἐκεῖνο, καὶ ἦν εἰς τὸ παρελθόν. εἰ δὲ μηδεὶς χρόνος
time to that, and it was to the past. if but no one time
ὃς οὔ ποτε, πᾶς ἂν εἴη χρόνος πεπερασμένος.
which not at any time, all would be time defined.

ἆρ' οὖν ὑπολείψει; ἢ οὔ, εἴπερ αἰεὶ ἔστι κίνησις; ἄλλος οὖν ἢ
is it then will be left? or not, if ever always there is motion? another then or
ὁ αὐτὸς πολλάκις; δῆλον ὅτι ὡς ἂν ἡ κίνησις, οὕτω καὶ ὁ
the same often? it is clear that as ever the motion, so and the
χρόνος· εἰ μὲν γὰρ ἡ αὐτὴ καὶ μία γίγνεταί ποτε,
time. if on the one hand for the same and one it becomes at some time,
ἔσται καὶ χρόνος εἷς καὶ ὁ αὐτός, εἰ δὲ μή, οὐκ ἔσται.
it will be and time one and the same, if but not, it will not be.

ἐπεὶ δὲ τὸ νῦν τελευτὴ [222b] καὶ ἀρχὴ χρόνου, ἀλλ' οὐ τοῦ
since but the now is the end [222b] and the beginning of time, yet not of the
αὐτοῦ, ἀλλὰ τοῦ μὲν παρήκοντος τελευτή, ἀρχὴ δὲ τοῦ
same, but of the one that has passed end, the beginning but of the
μέλλοντος, ἔχοι ἂν ὥσπερ ὁ κύκλος ἐν τῷ αὐτῷ πως τὸ
future, it would hold if just as the circle in the same way the
κυρτὸν καὶ τὸ κοῖλον, οὕτως καὶ ὁ χρόνος ἀεὶ ἐν ἀρχῇ
convex and the concave, in this way and the time always in beginning
καὶ τελευτῇ. καὶ διὰ τοῦτο δοκεῖ ἀεὶ ἕτερος· οὐ γὰρ τοῦ
and end. and because of this it seems always different. not for of the
αὐτοῦ ἀρχὴ καὶ τελευτὴ τὸ νῦν· ἅμα γὰρ ἂν καὶ κατὰ τὸ
same beginning and end the now; together for would and according to the

αὐτὸ τἀναντία ἂν εἴη. καὶ οὐχ ὑπολείψει δή· αἰεὶ γὰρ ἐν

it the opposite if it should be. and not will remain indeed always for in

ἀρχῇ.

the beginning.

τὸ δ' ἤδη τὸ ἐγγύς ἐστι τοῦ παρόντος νῦν ἀτόμου μέρος τοῦ

the but already the near is of the present now atomic part of the

μέλλοντος χρόνου (πότε βαδίζεις; ἤδη, ὅτι ἐγγὺς ὁ χρόνος ἐν

future time (when will you walk? already, that near the time in

ᾧ μέλλει), καὶ τοῦ παρεληλυθότος χρόνου τὸ μὴ πόρρω τοῦ

which is about to), and of the past time the not far from the

νῦν (πότε βαδίζεις; ἤδη βεβάδικα). τὸ δὲ Ἴλιον φάναι ἤδη

now (when will you walk? already you have walked). the but Ilium to say already

ἐαλωκέναι οὐ λέγομεν, ὅτι λίαν πόρρω τοῦ νῦν. καὶ τὸ ἄρτι

to have been captured not we say, that very far from the now. And the just

τὸ ἐγγὺς τοῦ παρόντος νῦν [τὸ] μόριον τοῦ παρελθόντος. πότε

the near of the present now [the] part of the past. When

ἦλθες; ἄρτι, ἐὰν ᾖ ὁ χρόνος ἐγγὺς τοῦ ἐνεστῶτος νῦν.

did you arrive? just, if it is the time near of the present now.

πάλαι δὲ τὸ πόρρω. τὸ δ' ἐξαίφνης τὸ ἐν ἀναισθήτῳ χρόνῳ διὰ

Long ago but the far. the but suddenly the in an unperceived time through

μικρότητα ἐκστάν· μεταβολὴ δὲ πᾶσα φύσει ἐκστατικόν. ἐν δὲ

smallness becoming ecstatic; change and every nature ecstatic. In and

τῷ χρόνῳ πάντα γίγνεται καὶ φθείρεται· διὸ καὶ οἱ μὲν

the time everything comes to be and is corrupted; therefore and the on one hand,

σοφώτατον ἔλεγον, ὁ δὲ Πυθαγόρειος Πάρων ἀμαθέστατον,

the wisest said, the but Pythagorean Parmenides most ignorant,

ὅτι καὶ ἐπιλανθάνονται ἐν τούτῳ, λέγων ὀρθότερον. δῆλον

that also they forget in this, saying more correctly. It is clear

οὖν ὅτι φθορᾶς μᾶλλον ἔσται καθ' αὑτὸν αἴτιος

therefore that of destruction more there will be throughout itself responsible

ἢ γενέσεως, καθάπερ ἐλέχθη καὶ πρότερον (ἐκστατικὸν γὰρ

than of generation, just as has been said also previously (the ecstatic for

ἡ μεταβολὴ καθ' αὑτήν), γενέσεως δὲ καὶ τοῦ εἶναι κατὰ

the change in itself), of generation but also of the being according to

συμβεβηκός. σημεῖον δὲ ἱκανὸν ὅτι γίγνεται μὲν οὐδὲν

accident. A sign but sufficient that it comes to be on one hand nothing

ἄνευ τοῦ κινεῖσθαί πως αὐτὸ καὶ πράττειν, φθείρεται δὲ καὶ

without the to be moved somehow itself and to act, is corrupted. but and

μηδὲν κινούμενον. καὶ ταύτην μάλιστα λέγειν εἰώθαμεν ὑπὸ τοῦ

nothing moving. and this especially to say we are accustomed under the

χρόνου φθοράν. οὐ μὴν ἀλλ' οὐδὲ ταύτην ὁ χρόνος ποιεῖ, ἀλλὰ

time decay. not indeed but nor this the time makes, but

συμβαίνει ἐν χρόνῳ γίγνεσθαι καὶ ταύτην τὴν μεταβολήν.

it happens in time to become and this the change.

ὅτι μὲν οὖν ἔστιν ὁ χρόνος καὶ τί, καὶ ποσαχῶς

that on the one hand therefore is the time and what, and in what ways

λέγεται τὸ νῦν, καὶ τί τὸ ποτὲ καὶ τὸ ἄρτι καὶ τὸ ἤδη καὶ τὸ

is said the now, and what the ever and the just and the already and the

πάλαι καὶ τὸ ἐξαίφνης, εἴρηται.

long ago and the suddenly, it has been said.

Κεφάλαιο 14

Chapter 14

Τούτων δ' ἡμῖν οὕτω διωρισμένων φανερὸν ὅτι πᾶσα μεταβολὴ

Of these and to us thus having been defined obvious that every change

καὶ ἅπαν τὸ κινούμενον ἐν χρόνῳ. τὸ γὰρ θᾶττον καὶ βραδύτερον

and all the moving in time. the for faster and slower

κατὰ πᾶσάν ἐστιν μεταβολήν (ἐν πᾶσι γὰρ οὕτω φαίνεται)·

according to every is change (in all for thus it appears).

λέγω δὲ θᾶττον κινεῖσθαι τὸ πρότερον [223a] μεταβάλλον εἰς

I say but faster to be moving the earlier [223a] changing into

τὸ ὑποκείμενον κατὰ τὸ αὐτὸ διάστημα καὶ ὁμαλὴν κίνησιν

the substrate according to the same interval and uniform motion

κινούμενον (οἷον ἐπὶ τῆς φορᾶς, εἰ ἄμφω κατὰ τὴν περιφερῆ

moving (like in the motion, if both according to the circular

κινεῖται ἢ ἄμφω κατὰ τὴν εὐθεῖαν· ὁμοίως δὲ καὶ ἐπὶ τῶν

it moves or both according to the straight line. Similarly but also for the

ἄλλων). ἀλλὰ μὴν τό γε πρότερον ἐν χρόνῳ ἐστί· πρότερον

others). But indeed the certainly earlier in time it is. Earlier

γὰρ καὶ ὕστερον λέγομεν κατὰ τὴν πρὸς τὸ νῦν ἀπόστασιν,

for and later we say according to the towards the now distance,

τὸ δὲ νῦν ὅρος τοῦ παρήκοντος καὶ τοῦ μέλλοντος· ὥστ'

the but now boundary of the past and of the future; so that

ἐπεὶ τὰ νῦν ἐν χρόνῳ, καὶ τὸ πρότερον καὶ ὕστερον ἐν χρόνῳ

since the now in time, and the earlier and later in time

ἔσται· ἐν ᾧ γὰρ τὸ νῦν, καὶ ἡ τοῦ νῦν ἀπόστασις. (ἐναντίως

it will be; in which for the now, and the of the now distance. (conversely

δὲ λέγεται τὸ πρότερον κατά τε τὸν παρεληλυθότα χρόνον καὶ

but it is said the earlier according to both the past time and

τὸν μέλλοντα· ἐν μὲν γὰρ τῷ παρεληλυθότι πρότερον λέγομεν τὸ
the future; in indeed for the past earlier we say the
πορρώτερον τοῦ νῦν, ὕστερον δὲ τὸ ἐγγύτερον, ἐν δὲ τῷ μέλλοντι
farther of the now, later but the nearer, in and the future
πρότερον μὲν τὸ ἐγγύτερον, ὕστερον δὲ τὸ πορρώτερον.) ὥστε
earlier indeed the nearer, later but the farther.) so
ἐπεὶ τὸ μὲν πρότερον ἐν χρόνῳ, πάσῃ δ' ἀκολουθεῖ κινήσει
since the on one hand earlier in time, in all but follows motion
τὸ πρότερον, φανερὸν ὅτι πᾶσα μεταβολὴ καὶ πᾶσα κίνησις ἐν
the earlier, evident that every change and every motion in
χρόνῳ ἐστίν. ἄξιον δ' ἐπισκέψεως καὶ πῶς ποτε ἔχει ὁ
time is. worthy but of consideration and how at any time is related the
χρόνος πρὸς τὴν ψυχήν, καὶ διὰ τί ἐν παντὶ δοκεῖ εἶναι ὁ
time to the soul, and through what in everything it seems to be the
χρόνος, καὶ ἐν γῇ καὶ ἐν θαλάττῃ καὶ ἐν οὐρανῷ. ἢ ὅτι κινήσεώς
time, and in earth and in sea and in heaven. Or that of motion
τι πάθος ἢ ἕξις, ἀριθμός γε ὤν, ταῦτα δὲ κινητὰ πάντα (ἐν
any passion or habit, number indeed being, these but are movable all (in
τόπῳ γὰρ πάντα), ὁ δὲ χρόνος καὶ ἡ κίνησις ἅμα κατά τε
place for all), the but time and the motion together according to and
δύναμιν καὶ κατ' ἐνέργειαν;
power and according to action

πότερον δὲ μὴ οὔσης ψυχῆς εἴη ἂν ὁ χρόνος ἢ οὔ, ἀπορήσειεν
whether but not being of a soul would be might the time or not, would wonder
ἄν τις. ἀδυνάτου γὰρ ὄντος εἶναι τοῦ ἀριθμήσοντος
might someone. of the impossible for being to be of the enumerated

ἀδύνατον καὶ ἀριθμητόν τι εἶναι, ὥστε δῆλον ὅτι οὐδ'
impossible and countable something to be, so that it is clear that neither
ἀριθμός. ἀριθμὸς γὰρ ἢ τὸ ἠριθμημένον ἢ τὸ ἀριθμητόν. εἰ δὲ
number. number for or the counted or the countable. if but
μηδὲν ἄλλο πέφυκεν ἀριθμεῖν ἢ ψυχὴ καὶ ψυχῆς νοῦς, ἀδύνατον
nothing else has become to be counted or a soul and of the soul mind, impossible
εἶναι χρόνον ψυχῆς μὴ οὔσης, ἀλλ' ἢ τοῦτο ὅ ποτε ὂν
to be time of the soul not being, but or this that at some time being
ἔστιν ὁ χρόνος, οἷον εἰ ἐνδέχεται κίνησιν εἶναι ἄνευ ψυχῆς. τὸ
is the time, such as if it is possible motion to be without a soul. the
δὲ πρότερον καὶ ὕστερον ἐν κινήσει ἐστίν· χρόνος δὲ ταῦτ' ἐστὶν
but earlier and later in motion it is: time but these is
ᾗ ἀριθμητά ἐστιν.
in which measurable exists.

ἀπορήσειε δ' ἄν τις καὶ ποίας κινήσεως ὁ χρόνος ἀριθμός.
it would wonder but if anyone and what kind of motion the time number.
ἢ ὁποιασοῦν; καὶ γὰρ γίγνεται ἐν χρόνῳ καὶ φθείρεται καὶ
or whatever kind? and for it becomes in time and it is destroyed and
αὐξάνεται καὶ ἀλλοιοῦται καὶ φέρεται· ᾗ οὖν κίνησίς
it increases and it is altered and it is carried; in which therefore motion
ἐστι, ταύτῃ ἐστὶν ἑκάστης κινήσεως ἀριθμός. διὸ κινήσεώς ἐστιν
is, in this is each motion's number. therefore of motion is
ἁπλῶς ἀριθμὸς [223b] συνεχοῦς, ἀλλ' οὐ τινός.
simply a number [223b] of continuous, but not of any.

ἀλλ' ἔστι νῦν κεκινῆσθαι καὶ ἄλλο· ὧν ἑκατέρας τῆς
but is now to be set in motion and another; of which of each of the

κινήσεως εἴη ἂν ἀριθμός. ἕτερος οὖν χρόνος ἔστιν, καὶ

movement may be if a number. another therefore time is, and

ἅμα δύο ἴσοι χρόνοι ἂν εἶεν· ἢ οὔ; ὁ αὐτὸς γὰρ

at the same time two equal times if they may be; or not? the same for

χρόνος καὶ εἷς ὁ ἴσος καὶ ἅμα· εἴδει δὲ καὶ οἱ μὴ

time and one the equal and at the same time; in kind but also the not

ἅμα· εἰ γὰρ εἶεν κύνες, οἱ δ' ἵπποι, ἑκάτεροι δ' ἑπτά,

at the same time; if for they were dogs, the but horses, each but seven,

ὁ αὐτὸς ἀριθμός. οὕτω δὲ καὶ τῶν κινήσεων τῶν ἅμα

the same number. thus but also of the movements of the at the same time

περαινομένων ὁ αὐτὸς χρόνος, ἀλλ' ἡ μὲν ταχεῖα ἴσως ἡ

completing the same time, but the indeed swift perhaps the

δ' οὔ, καὶ ἡ μὲν φορὰ ἡ δ' ἀλλοίωσις· ὁ μέντοι χρόνος

but not, and the indeed motion the but alteration; the however time

ὁ αὐτός, εἴπερ καὶ [ὁ ἀριθμὸς] ἴσος καὶ ἅμα, τῆς

the same, if indeed and the number] equal and at the same time, of the

τε ἀλλοιώσεως καὶ τῆς φορᾶς. καὶ διὰ τοῦτο αἱ μὲν

and change and of the motion. and through this the on one hand

κινήσεις ἕτεραι καὶ χωρίς, ὁ δὲ χρόνος πανταχοῦ

movements are different and separate, the on the other hand time everywhere

ὁ αὐτός, ὅτι καὶ ὁ ἀριθμὸς εἷς καὶ ὁ αὐτὸς πανταχοῦ ὁ

the same, because and the number one and the same everywhere the

τῶν ἴσων καὶ ἅμα. ἐπεὶ δ' ἔστι φορὰ καὶ ταύτης

of the equals and at the same time. since indeed there is a period and of this

ἡ κύκλῳ, ἀριθμεῖται δ' ἕκαστον ἑνί τινι συγγενεῖ, μονάδες

the cycle, it is counted indeed each one certain kindred, units

μονάδι, ἵπποι δ' ἵππῳ, οὕτω <δὲ> καὶ ὁ χρόνος χρόνῳ τινὶ

unit, horses indeed horse, thus <and> and the time to time a certain

ὡρισμένῳ, μετρεῖται δ', ὥσπερ εἴπομεν, ὅ τε χρόνος κινήσει

defined, it is measured indeed, just as we said, that and time motion

καὶ ἡ κίνησις χρόνῳ (τοῦτο δ' ἐστίν, ὅτι ὑπὸ τῆς ὡρισμένης

and the movement in time (this but it is, because by the defined

κινήσεως χρόνῳ μετρεῖται τῆς τε κινήσεως τὸ ποσὸν καὶ τοῦ

motion in time is measured the and motion the quantity and of the

χρόνου)-εἰ οὖν τὸ πρῶτον μέτρον πάντων τῶν συγγενῶν,

time) - is if therefore the first measure of all the

ἡ κυκλοφορία ἡ ὁμαλὴς μέτρον μάλιστα, ὅτι ὁ

related, the circulation the smooth measure especially, because

ἀριθμὸς ὁ ταύτης γνωριμώτατος. ἀλλοίωσις μὲν οὖν οὐδὲ

the number the of this most known. change indeed therefore

αὔξησις οὐδὲ γένεσις οὐκ εἰσὶν ὁμαλεῖς, φορὰ δ' ἔστιν. διὸ

nor increase nor generation are not included smooth, motion but is.

καὶ δοκεῖ ὁ χρόνος εἶναι ἡ τῆς σφαίρας κίνησις, ὅτι

therefore and it seems the time to be the of the sphere motion,

ταύτῃ μετροῦνται αἱ ἄλλαι κινήσεις καὶ ὁ χρόνος ταύτῃ

because by this are measured the other motions and the time

τῇ κινήσει.

by this the

διὰ δὲ τοῦτο καὶ τὸ εἰωθὸς λέγεσθαι συμβαίνει· φασὶν γὰρ

through but this and the accustomed to be said happens; they say for

κύκλον εἶναι τὰ ἀνθρώπινα πράγματα, καὶ τῶν ἄλλων τῶν κίνησιν

a circle to be the human things, and of the others of the motions

ἐχόντων φυσικὴν καὶ γένεσιν καὶ φθοράν. τοῦτο δέ, ὅτι ταῦτα

having physical and generation and decay. this but, that these

πάντα τῷ χρόνῳ κρίνεται, καὶ λαμβάνει τελευτὴν καὶ ἀρχὴν
all by the time is judged, and takes an end and a beginning
ὥσπερ ἂν εἰ κατά τινα περίοδον. καὶ γὰρ ὁ χρόνος αὐτὸς
just as if it is according to a certain period. And for the time itself
εἶναι δοκεῖ κύκλος τις· τοῦτο δὲ πάλιν δοκεῖ, διότι τοιαύτης
to be seems to be a circle some; this but again seems, because of such
ἐστὶ φορᾶς μέτρον καὶ μετρεῖται αὐτὸς ὑπὸ τοιαύτης. ὥστε τὸ
is a burden a measure and is measured itself by such. So that the
λέγειν εἶναι τὰ γιγνόμενα τῶν πραγμάτων κύκλον τὸ λέγειν ἐστὶν
saying to be the generated of the things a circle the saying is
τοῦ χρόνου εἶναί τινα κύκλον· τοῦτο δέ, ὅτι μετρεῖται τῇ
of the of time to be some circle; this but, that is measured by the
κυκλοφορίᾳ· παρὰ γὰρ τὸ μέτρον οὐδὲν [224a] ἄλλο παρεμφαίνεται
circulation; for for the measure nothing [224a] else appears
τῷ μετρουμένῳ, ἀλλ' ἢ πλείω μέτρα τὸ ὅλον.
to the measured, but rather more measures the whole.

λέγεται δὲ ὀρθῶς καὶ ὅτι ἀριθμὸς μὲν ὁ αὐτὸς ὁ τῶν
is said and correctly and that number on one hand the same the of the
προβάτων καὶ τῶν κυνῶν, εἰ ἴσος ἑκάτερος, δεκὰς δὲ οὐχ ἡ αὐτὴ
sheep and of the dogs, if equal each, tens and not the same
οὐδὲ δέκα τὰ αὐτά, ὥσπερ οὐδὲ τρίγωνα τὰ αὐτὰ τὸ ἰσόπλευρον καὶ
nor ten the same, just as nor triangles the same the equilateral and
τὸ σκαληνές, καίτοι σχῆμά γε ταὐτό, ὅτι τρίγωνα ἄμφω· ταὐτὸ
the scalene, and yet shape indeed the same, that triangles both. the same
γὰρ λέγεται οὗ μὴ διαφέρει διαφορᾷ, ἀλλ' οὐχὶ οὗ διαφέρει,
for is said of which not differs in difference, but not of which it differs,

οἷον τρίγωνον τριγώνου <τριγώνου> διαφορᾷ διαφέρει·

such as triangle of a triangle <of a triangle> in difference it differs.

τοιγαροῦν ἕτερα τρίγωνα· σχήματος δὲ οὔ, ἀλλ' ἐν τῇ αὐτῇ

therefore other triangles. of shape and no, but in the same

διαιρέσει καὶ μιᾷ. σχῆμα γὰρ τὸ μὲν τοιόνδε κύκλος, τὸ δὲ

division and one. shape for the on one hand such circle, the but

τοιόνδε τρίγωνον, τούτου δὲ τὸ μὲν τοι όνδε ἰσόπλευρον,

such triangle, of this but the on the one hand such being equilateral,

τὸ δὲ τοιόνδε σκαληνές. σχῆμα μὲν οὖν τὸ αὐτό, καὶ τοῦτο

the but such scalene. shape on one hand therefore the same, and this

τρίγωνον, τρίγωνον δ' οὐ τὸ αὐτό. καὶ ἀριθμὸς δὴ ὁ αὐτός (οὐ

triangle, triangle but not the same. and number indeed the same (not

γὰρ διαφέρει ἀριθμοῦ διαφορᾷ ὁ ἀριθμὸς αὐτῶν), δεκὰς δ' οὐχ ἡ

for differs in number aspects the number of them), tens but not the

αὐτή· ἐφ' ὧν γὰρ λέγεται, διαφέρει· τὰ μὲν γὰρ κύνες,

same; concerning which for it is said, differs. the on one hand for dogs,

τὰ δ' ἵπποι.

the but horses.

καὶ περὶ μὲν χρόνου καὶ αὐτοῦ καὶ τῶν περὶ αὐτὸν

and about on one hand time and itself and of the concerning it

οἰκείων τῇ σκέψει εἴρηται.

of relatives in the discussion it has been said.